SIEMENS Healthineers

MAGNETOM

Troubleshooting Guide System

Patient Handling

Aera
Skyra
Avanto fit
Skyra fit
Prisma
Prisma fit

Document Version

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1.1 Notes

After switching-off the power for the component, perform "Reboot Meas & Recon".

1.1.1 Safety Information and/or References to Safety Information

Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-specific safety information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the safety notes may result in death or serious injury.

» **Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.**

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Move the table out of the magnet room, maintain a safe distance to the magnetic field, or ramp down the magnet.**

Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance <= 100 mOhm with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "subcomponent" (e.g. direct position drive) and the protective earth wire connected to the "main component" (e.g. patient table).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Always fasten the safety bolts in the storage position after use !

1.1.2 Product-specific Remarks

Tab. 1 Product-specific Remarks 3 T

System	Skyra / Skyra^{fit} / Prisma / Prisma^{fit}	Skyra / Skyra^{fit} / Prisma / Prisma^{fit}
Patient table- Type	Fixed table	Moveable table
Material number	104 33 196 K2302	104 33 198 K2303

Tab. 2 Product-specific Remarks 1.5 T

System	Aera / Avanto^{fit} / Avanto fit ex-factory	Aera / Avanto^{fit} / Avanto fit ex-factory
Patient table- Type	Fixed table	Moveable table
Material number	104 33 195 K2302	104 33 197 K2303

1.2 Requirements

1.2.1 Additionally Required Documents

Function Description "Patient Handling" chapter (➔ Interactive Functional Description / M7-000.850.01).

Replacement of Parts "Patient Handling" chapter (➔ Patient Handling / M7-000.841.15).

1.2.2 Required Aids and Tools

- Non-magnetic steel tool set (incl. allen keys with ball head).
- Digital voltmeter (material number 97 03 976) or equivalent.
- Ground test meter.

2.1 Troubleshooting Strategy

This section of the TSG deals with problems regarding the PMU. The strategies and procedures provide the necessary information for servicing the PMU. An advanced trouble shooting procedure is included in this document.

The primary components of the **PMU** Physiological Measurement Unit are:

- **PDAU**-Physiological Data Acquisition unit, included on the RFCEL_COM board
- **PERU**-Physiological ECG and Respiratory Sensor Unit
- **PPU**-Peripheral Pulse Unit

The physiological signals are shown on the Patient Data Display at the magnet and on the physiological display UI.

> The PERU and the PPU are supplied with a small white connector. This connector prevents the rechargeable battery from discharging. Disconnect the connectors if the units are in operation.

The primary tool for PMU troubleshooting is the **PMU** test, which is available in the Service Software platform under **Test Tools**. The error messages are shown in the event log. The following table is designed to provide the CSE with the error symptom and identify the specified FRU in the PMU system. **First** use the **Test Tools** before starting any other service step.

The "Replacement of Parts" folder lists the necessary adjustments and the final checks when replacing the FRU.

> The PERU and PPU continue to transmit only inside the magnetic field.

Regarding the revision states for PERU/PPU listed below, they transmit only within the magnetic field as compared to previous conditions. This may lead to confusion at the customer sites. Erroneously, the customers may think that the component no longer function correctly once it is outside the magnetic field.

PMU_098

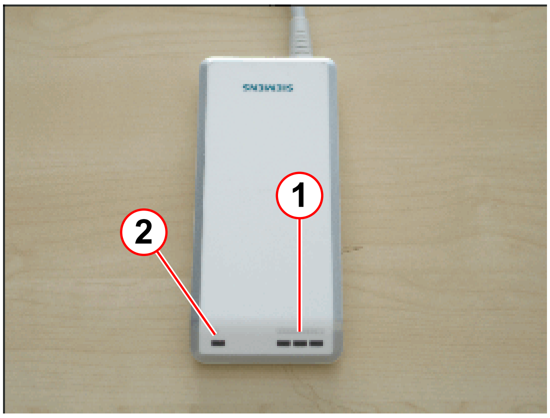
PERU_098 beginning with Rev. 03

PPU_098 beginning with Rev. 02

2.2 LEDs, Test Points, Switches

2.2.1 LEDs and Displays

Fig. 1: LED indication



(1) Battery status (green)
(2) Application error ecg or pulse sensor (red)

2.2.2 Test Points

n.a.

2.2.3 Service Switches

n.a.

2.2.4 Jumpers

n.a.

2.3 Status Information, Tests

Advanced trouble shooting procedure for VCG triggering			
Symptom	**Possible root cause**	**Probable cause**	**Instructions**
Zero line at the PhysioDisplay and the PMU Display	One or more electrodes do not form appropriate contact with the patient's skin - "Electrodes not properly attached" is reported in the status line of the PhysioDisplay - see (➔ Fig. 2 Page 12)	Dry electrodes	Refer to (➔ Symptom: Zero Line at the PhysioDisplay and the PMU Display / Page 12)
		Incorrect electrode type	Refer to (➔ Symptom: Zero Line at the PhysioDisplay and the PMU Display / Page 12)
	Interference by a second PERU (➔ Fig. 3 Page 12)	another unused PERU	Refer to (➔ Symptom: Zero Line at the PhysioDisplay and the PMU Display / Page 12)
	No signal received by PERU - "No Signal from PERU" is reported in the status line of the PhysioDisplay - see (➔ Fig. 4 Page 12)	The patient has **placed his hand on the transmitter box**	Refer to (➔ Symptom: Zero Line at the PhysioDisplay and the PMU Display / Page 12)
		The **battery is low**	Refer to (➔ Symptom: Zero Line at the PhysioDisplay and the PMU Display / Page 12)
Intermittent zero line at the PhysioDisplay and at the PMU Display	Interference by a second PERU	another unused PERU	Refer to (➔ Symptom: Intermittent zero line at the PhysioDisplay and at the PMU Display / Page 13)

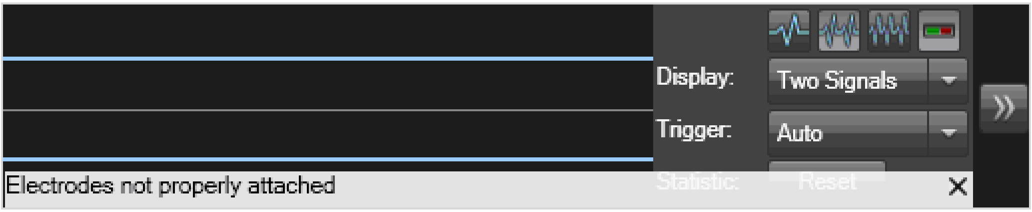
Advanced trouble shooting procedure for VCG triggering			
Symptom	**Possible root cause**	**Probable cause**	**Instructions**
Zero line at the PhysioDisplay, but correct signal at the PDD in the Examination Room	"No valid status from PDAU available" is reported in the status line of the Physio-Display, see - (→ Fig. 6 Page 13)	Software (e.g. ExamTaskCard --> Restart)	Refer to (→ Symptom: Zero Line at the PhysioDisplay, but Correct Signal at the PMU Display in the Examination Room / Page 13)
Triggering unreliable/ no triggering	Refer to (→ Symptom: Triggering unreliable/ No Triggering / Page 14)	(→ Symptom: Triggering unreliable/ No Triggering / Page 14)	(→ Symptom: Triggering unreliable/ No Triggering / Page 14)

2.3.1 Symptom: Zero Line at the PhysioDisplay and the PMU Display

2.3.1.1 Possible Root Causes

- One or more electrodes do not form appropriate contact with the patient's skin - "Electrodes not properly attached" is reported in the status line of the PhysioDisplay - see

Fig. 2: Screenshot "Electrodes not properly attached"



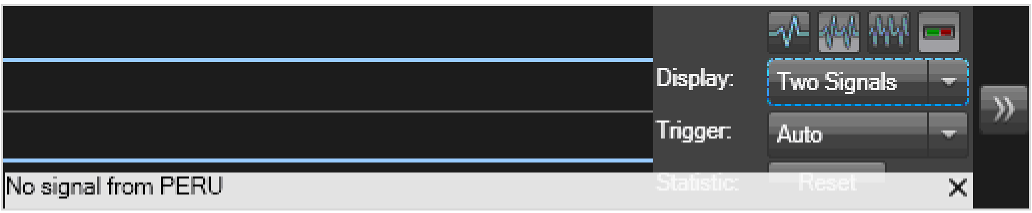
  - **Due to dry electrodes** - check the expiry date
  - **Due to an incorrect electrode type** - we highly recommend the Conmed 2700 Cleartrace electrodes, as they ensure reliable skin contact even for very time-consuming examinations. Other electrodes may tend to loose contact over time, especially for stress examinations, where the patients may tend to sweat. Due to **improper preparation** of the patient's **skin**: prepare the skin by means of NuPrep Skin Prepping Gel (please avoid alcohol, which removes the electrolytes from the skin), shave the patient's skin in the areas intended for the electrode placement (if necessary).
- **Interference by a second PERU**

Fig. 3: Buckler of PERU

There is **another (unused) PERU** in the examination room or outside of it with the door open (e.g. not properly inserted into the charger), **which reports an ECG electrode fault** - see (→ Fig. 2 Page 12). Place all unused PERUs in their chargers or plug in the buckler .

- No signal received by PERU - "No Signal from PERU" is reported in the status line of the PhysioDisplay - see (→ Fig. 4 Page 12)

Fig. 4: Screenshot "No signal from PERU"



Possible root cause

- The patient has **placed his hand on the transmitter box** (the larger of the 2 boxes), which leads to an excessive signal attenuation.
- The **battery is low** (none of the LEDs at the unit are flashing - prior to this behavior, only one green LED starts flashing and a weak battery is reported in the PhysioDisplay (red bar in the battery symbol) - in this condition, there is at least 1 more hour of operation before the unit needs recharging).

2.3.2 Symptom: Intermittent zero line at the PhysioDisplay and at the PMU Display

Intermittent zero lines with an "ECG Electrode fault" reported in the status line of the Physio Display.

Fig. 5: Intermittend Zeroline

2.3.2.1 Possible Root Cause

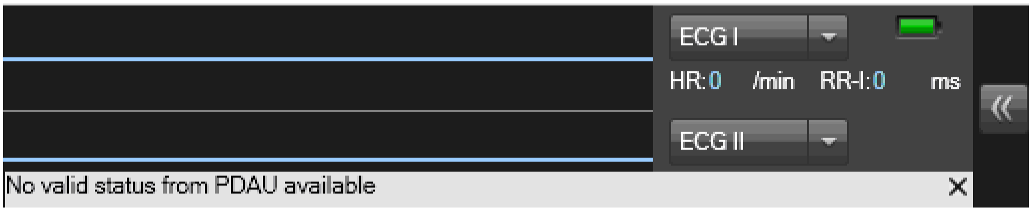
- **Interference by a second PERU**

  There is **another (unused) PERU** in the examination room or outside of it with the door open, **which reports the "ECG electrode fault"**. Please place all unused PERUs accurately into their chargers (Check LEDs. They indicate, the unit is charged or charging is in progress).

2.3.3 Symptom: Zero Line at the PhysioDisplay, but Correct Signal at the PMU Display in the Examination Room

"No valid status from PDAU available" is reported in the status line of the PhysioDisplay, see - (➔ Fig. 6 Page 13)

Fig. 6: Screenshot "No valid status from PDAU available"



Do not replace the PERU - it is fine in this case.

2.3.3.1 Possible root Causes

Connection MARS-RFCEL lost.

2.3.4 Symptom: Triggering unreliable/ No Triggering

From experience, most of the problems with VCG gating relate to disadvantageous patient handling during the learning phase, which can be quite easily improved for most of the examinations.

Check the most important notes below.

2.3.4.1 Application Note for Stable VCG Triggering

To ensure reliable operation of the VCG triggering it is absolutely essential, that the following notes are observed.

- After attaching the PERU as described in the System Manual, the 3 leads (I,II and III) are evaluated. The R-waves should be clearly recognizable for all leads and as a result suitable for triggering during the examination.
- After positioning the Body Matrix, wait for the learning phase to finish before moving the patient into the magnet (in table home or near home position). When the learning phase has finished, the text "..." in the status line of the PDD/ Physio page is disappearing

2.3.4.2 Detailed Information

1. The learning phase of the VCG is complete, when the patient table is located within 300 mm (118 inches) from the home position (learning range). The home position works best due to the lowest B0 stray field, which ensures T waves are minimally elevated by the magnetohydrodynamic effect.
2. The learning phase is interrupted as soon as the table is moved.
3. If the table is stopped within the learning range, the learning phase is continued.
4. At least 3 regular heartbeats are needed for the learning phase to finish successfully (regular means 3 heartbeats acquired with the patient lying still **and** the patient table remains at the same position within the learning range). After a first set of 3 **regular** heartbeats has been learned, the learning phase runs further till the table is moved beyond the learning range
5. If the table is moved more than 300 mm away from home position, the learning phase is terminated.
6. Another (new) learning phase can be started with the patient in the magnet by means of the Relearn option in the Physio Display menu (opens up by a right mouse click on the UI, only available for the "Auto" trigger choice).
7. If the PERU is connected to the attached electrodes **after** the patient table has been already moved into the magnet **beyond the learning range**, no triggers can be gen-

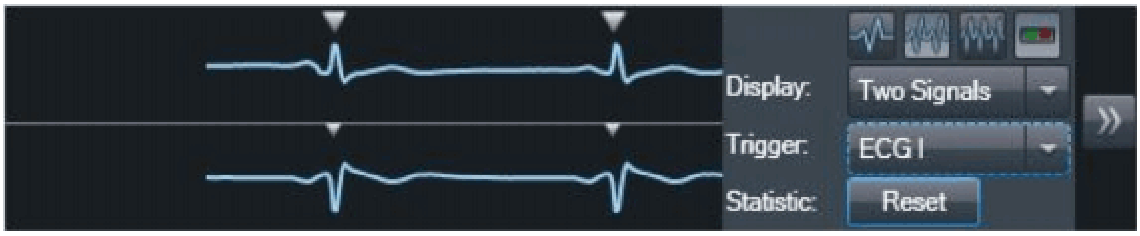
erated by the VCG algorithm due to the lack of learning data and therefore the **red trigger triangles** in the Physio Display are **missing**.

Fig. 7: No trigger pulses

Moving the Patient Table out to the home position solves the problem because the learning phase is then initially started.

If it is too disadvantageous to move the patient table back to home position, switching to the triggering modes "ECG I", "ECG II" or "ECG III" at the PhysioDisplay circumvents this problem, as those 2 modes don't use the learning data (please observe the notes regarding the 1-channel triggering modes below in this special case).

Fig. 8: Trigger pulses channel I



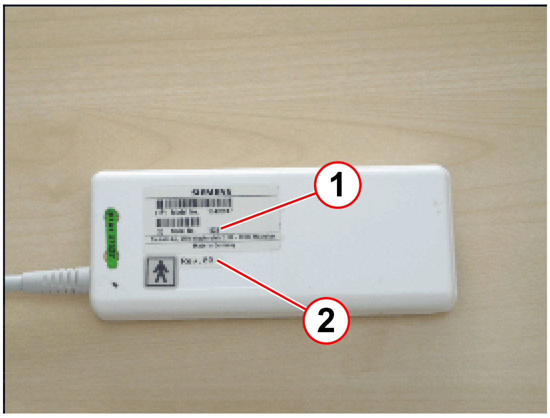
8. In the default trigger mode "Auto" the best 2 of the available 3 ECG channels are used for triggering. It is highly recommended, to commonly use this mode. It also works fine for most of the examinations, if one of the ECG channels is significantly smaller than the other ones.

> A display of all 3 ECG channels in the PhysioDisplay simplifies this assessment.

If no R waves are visible in one of the channels and the trigger mode “Auto” does not work correctly, it is recommended to use one of the other channels for triggering (modes "ECG I", "ECG II" or "ECG III"), or try to optimize the electrode placement to get a more distinct R wave in the weaker channel.

2.3.5 Problems, which cannot be resolved with the above Informations

Fig. 9: Serial number, revision level



(1) Serial number
(2) Revision level

In case all mentioned tips can not help you resolving a VCG triggering problem, it is very helpful for us, if you could give us the serial number and the hardware revision level of the PERU (see (➔ Fig. 9 Page 16)).

Additionally, it helps us for further problem diagnosis, if you could send us ECG logfiles (please refer to the instructions below.

2.3.5.1 Recording of Physiological Signals

- The resulting logfiles can be found in the directory c:\MedCom\log

  The extension of the ECG files is .ecg
- The necessary inputs are indicated italic

Open Windows XP Start Menu:

Menu item "Run..."

Open: ideacmdtool

The ideacmdtool menu is entered...

Start/ Stop the signal logging (e.g. for the ECG signal):

Remark: all inputs need to be confirmed by the Enter (CR) Key.

```
-----------------------------------------------------------------
IDEACmdTool Menu
-----------------------------------------------------------------
1. PMU control...
2. PAS unload
3. Measure
4. Online Export Defaults
5. Switches...
6. Set Default
```

```
------------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
------------------------------------------------------------------
Cmd >1

------------------------------------------------------------------
PMU control Menu
------------------------------------------------------------------
1. PMU Signal simulation
2. PMU Signal logging
------------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
------------------------------------------------------------------
PMU> 2

------------------------------------------------------------------
Menu
------------------------------------------------------------------
1. dump
2. setHWTraceLevel
3. isCallAllowed
4. configureIF
5. startHostLogECG
6. stopHostLogECG
7. startHostLogRespiration
8. stopHostLogRespiration
9. startHostLogPulse
10. stopHostLogPulse
11. startHostLogExternal
12. stopHostLogExternal
13. startHostLogAll
14. stopHostLogAll
------------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
------------------------------------------------------------------
Cmd > 5
enter filename: fl_trig140208
```

```
command startHostLogECG returned: kPerRetOK
```

Don't forget to Stop the logging after the session, please !

```
-----------------------------------------------------------------
Menu
-----------------------------------------------------------------
1. dump
2. setHWTraceLevel
3. isCallAllowed
4. configureIF
5. startHostLogECG
6. stopHostLogECG
7. startHostLogRespiration
8. stopHostLogRespiration
9. startHostLogPulse
10. stopHostLogPulse
11. startHostLogExternal
12. stopHostLogExternal
13. startHostLogAll
14. stopHostLogAll
-----------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
-----------------------------------------------------------------
Cmd > 6

command stopHostLogECG returned: kPerRetOK

-----------------------------------------------------------------
Menu
-----------------------------------------------------------------
1. dump
2. setHWTraceLevel
3. isCallAllowed
4. configureIF
5. startHostLogECG
6. stopHostLogECG
```

```
7. startHostLogRespiration
8. stopHostLogRespiration
9. startHostLogPulse
10. stopHostLogPulse
11. startHostLogExternal
12. stopHostLogExternal
13. startHostLogAll
14. stopHostLogAll
-----------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
-----------------------------------------------------------------
Cmd > q

-----------------------------------------------------------------
PMU control Menu
-----------------------------------------------------------------
1. PMU Signal simulation
2. PMU Signal logging
-----------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
-----------------------------------------------------------------
PMU> q

-----------------------------------------------------------------
IDEACmdTool Menu
-----------------------------------------------------------------
1. PMU control...
2. PAS unload
3. Measure
4. Online Export Defaults
5. Switches ...
6. Set Default
-----------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
-----------------------------------------------------------------
> q
```

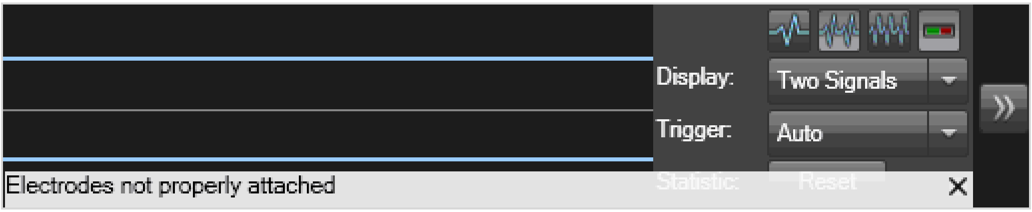
```
ideacmdtool exits...
```

2.3.6 Chart 01

The error messages shown under **symptom** occur in the physiological display UI.

Messages in the physiological display user interface (UI) about the PMU status; see example.

Fig. 10: Screenshot "Electrodes not properly attached"



Symptom	Probable Cause	Service Action	Instructions
01. No valid status from PDAU available	Connection RFCEL - MARS lost:	Check cable W17410 RFCEL_COM U1 ----> OPEC1/U1 Run SeSo TestTools: **PMU Test**, Test type: Comm/ Status and Interactive Test will fail; continue with next step.	n.a.
02. **No valid status from PDAU available**	PDAU supply voltage missing / RFCEL_COM supply voltages missing	Run SeSo TestTools: RFCEL_COM	n.a.
	RFCEL_COM defective		**Replacement of Parts** (→ RFCEL_COM Module / M7-000.841.14)
01. No valid status from PDAU available	OPEC1, power supply, jumper settings	Run SeSo TestTools: OPEC1	n.a.

2.3.7 Chart 02

If no signal is present on the physio or PMU Display at the magnet, review additional troubleshooting informations.			
Symptom	**Probable Cause**	**Service Action**	**Instructions**
No ECG signal is displayed on the physio display or PDD at the magnet	PERU, antenna, antenna cable, PDAU on RFCEL_COM	Run SeSo TestTools: **PMU Test**, Test type: Comm/ Status and Interactive (PERU positioned on table must be within the B0 field, electrode cables not connected to patient). Test o.k, test signal visible. (**Note**: Electrodes not properly attached and "No signal from PPU" message in the phys. display is normal.)	
	PERU, antenna, antenna cable, PDAU on RFCEL_COM	Position the PERU on the table (within the B0 field, electrode cables not connected to patient). Drive the table into the isocenter of magnet bore. Run SeSo TestTools: **PMU Test**, Test type: Comm/ Status and Interactive. Test o.k, test signal must be visible. (**Note**: Electrodes not properly attached and "No signal from PPU" message in the phys. display is normal.)	
	Electrode cables defective	Use a non-magnetic screwdriver to establish a short connection between all electrode clips. If the red LED (electrode fault) still blinks, replace the PERU	
	A second PERU is located somewhere in the examination room.	Remove the spare PERU from the examination room and insert it into the sensor charger.	
	Application problems (e.g., dry electrodes)	Refer to system manual; contact application specialist	System manual

If no signal is present on the physio or PMU Display at the magnet, review additional troubleshooting informations.			
Symptom	Probable Cause	Service Action	Instructions
No respiratory signal is displayed on the physio display or PDD at the magnet	PERU, antenna, antenna cable, PDAU on RFCEL_COM, respiratory cushion	Apply the respiratory cushion to your abdomen (refer to system manual). Check if the signal is displayed on the Physio Display or PDD at the magnet.	You may need a second person for help
	Respiratory cushion	Press the respiratory cushion with your fingers and check if there is air flow at the connector. If the previous tests are o.k. and you still have no respiratory signal, replace the PERU.	

If no signal is present on the physio or PMU Display at the magnet, review additional troubleshooting informations.

Symptom	Probable Cause	Service Action	Instructions
No pulse signal displayed in the physio display or PDD at the magnet	PPU, antenna, antenna cable, PDAU on RFCEL_COM	Run SeSo TestTools: **PMU Test**, Test type: Comm/Status and Interactive (PPU positioned on table, not connected to volunteer’s finger). Test o.k, test signal visible.	
	PPU, antenna, antenna cable, PDAU on RFCEL_COM	Prepare the PPU for pulse Triggering (PPU cables connected to the finger adapter). Position the PPU on the table. Drive the table into the isocenter of magnet bore. Run SeSo TestTools: **PMU Test**, Test type: Comm/Status and Interactive. Test o.k, test signal must be visible. (**Note**: "Finger clip not applied" message in the phys. display is normal.)	
	PPU	Position the PPU on the patient table. PMU Test, Test type: Comm/Status and Interactive test ok. Test signal visible. Attach the pulse sensor to your finger; check if the red LED (application fault) at the PPU stops flashing. Check if the pulse signal is displayed in the phys. display. If not o.k., replace the PPU.	
	A second PPU is located somewhere in the examination room.	Remove the spare PPU from the examination room and insert it into the sensor charger.	
	Application problems	Refer to system manual; contact application specialist	System manual

Fig. 11: Short-circuited PERU electrode ports

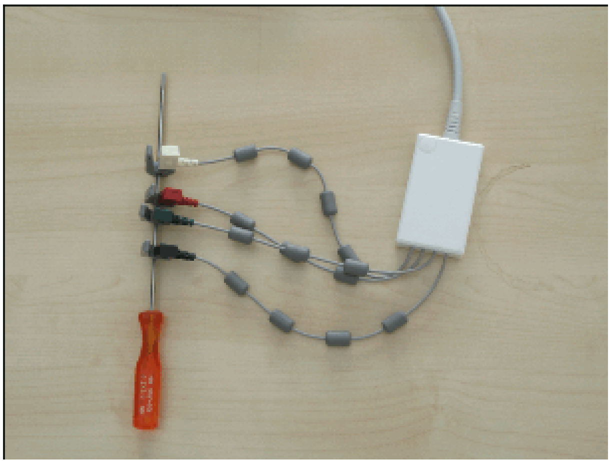


Fig. 12: PMU Interactive Screenshot

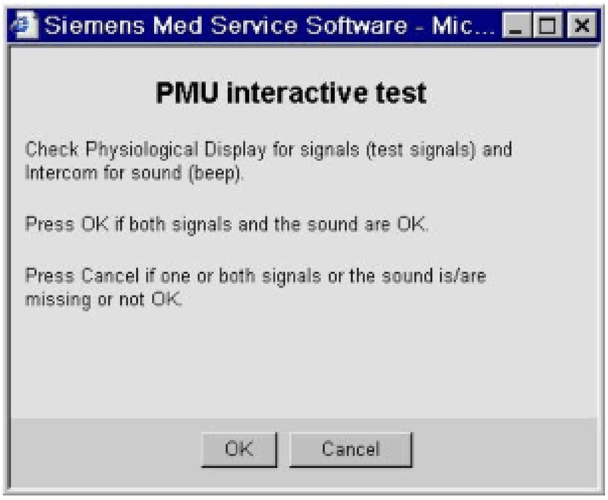


Fig. 13: PERU PPU Screenshot

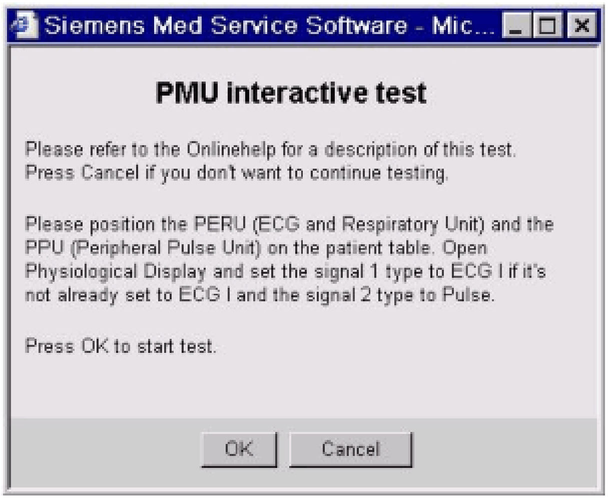
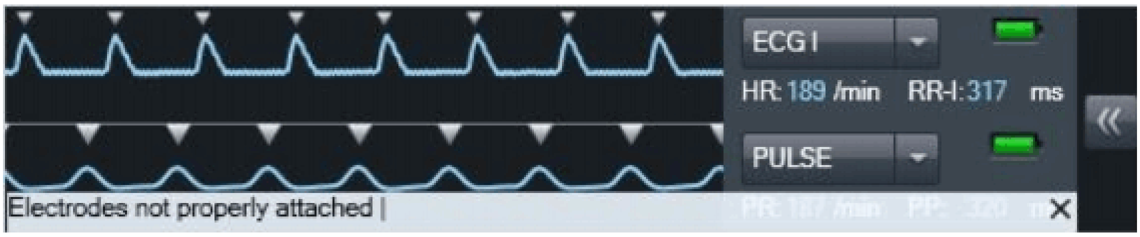


Fig. 14: PMU interactive test signal



2.4 Configuration

n.a.

3.1 Troubleshooting Strategy

This section of the TSG deals with problems regarding the PAMO 5. The strategies and procedures provide the necessary information for servicing the PAMO 5.

The primary components of the **PAMO 5** are:

- **PAMO 5** - Display
- **PAMO 5** - CCD Vario Camera
- **PAMO 5** - Combiner Set

The "Replacement of Parts" folder lists the necessary adjustments and the final checks when replacing the FRU.

Additional **troubleshooting information for the PDD** (DOT-Display) has been added to this chapter.

3.2 LEDs, Test Points

This chapter describe the LED´s and test points of the PAMO 5.

3.2.1 LEDs and Displays

Tab. 3 LED's

Location	name	Description
Back of the display	Status LED	ON, supply voltage is o.k
Camera front side	Status LED	ON. supply voltage is o.k

3.2.2 Test points

Tab. 4 Power supply voltage

Location	Name	Description
Camera	X22, Pin 2	P 16V
Camera	X22, Pin 3	GND
RFCEL/Intercom	X4, Pin 5	P 16V, camera 1 inside the RF-cabin
RFCEL/Intercom	X4, Pin 9	GND, camera 1 inside the RF-cabin
RFCEL/Intercom	X4, Pin 6	P 16V, camera 2 inside the RF-cabin
RFCEL/Intercom	X4, Pin 10	GND, camera 2 inside the RF-cabin
Filter plate/TX-box power supply	X25, Pin 3	P 16V, camera 1 outside the RF-cabin
Filter plate/TX-box power supply	X25, Pin 2	GND, camera 1 outside the RF-cabin
Filter plate/TX-box power supply	X26, Pin 3	P 16V, camera 2 outside the RF-cabin
Filter plate/TX-box power supply	X26, Pin 2	GND, camera 2 outside the RF-cabin

3.2.3 Service switches

n.a.

3.2.4 Jumpers

n.a.

3.3 Status Information, Tests

Tab. 5 Information on troubleshooting PAMO 5

Symptom	Probable Cause	Service action	Instructions
No Signal on display	No display supply voltage or defective display	Push the On/Off- button on the back of the display and check the LED, belonging to. If the LED is on, the supply voltage is o.k. Otherwise, there is no supply voltage or the display is defective.	n.a.
	No camera supply voltage or defective camera	Check the supply LED on the camera front side. It has to be on. The current supply voltage value can be checked at Camera/ X22/ pin 2 (P16V) and 3 (GND) (the cable has to be disconnected for this test). The value has to be between the limits 16,0V and 18,5V. Otherwise the supply source or the camera supply cable is defective. **Supply sources for cameras inside the RF Cabin:** Both cameras are supplied by a Y- cable adapter, which is connected to RFCEL/ Intercom/ X4. Pinning for Camera 1: Pin 5 (P16V) and Pin 9 (GND) Pinning for Camera 2: Pin 6 (P16V) and Pin 10 (GND) **Supply sources for cameras outside the RF Cabin:** Both cameras are supplied directly by the Filter Plate/TX-Box Power Supply/ X25 and X26. Pinning for Camera 1: X25/ Pin 3 (P16V) and Pin 2 (GND) Pinning for Camera 2: X26/ Pin 3 (P16V) and Pin 2 (GND)	n.a.

Symptom	Probable Cause	Service action	Instructions
No Signal on display	Defective FOC between camera and display	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	No combiner supply voltage or defective combiner (if a combiner is used)	Connect the FOC from one camera directly to the display (not via combiner). If the signal is visible on the screen, camera and display work properly. Otherwise proceed Service actions above. Check both cables between combiner and display (supply voltage and video signal) due to correct connection. If the problem still persists replace the combiner.	n.a.
Image on the display is dark and/ or noisy	Lens aperture of the camera is not open completely. Lens or camera is defective.	Open the lens aperture of the camera lens completely. If the problem still persists, replace the camera lens or camera. If the problem is ongoing, replace the display also.	n.a.
Interferences on the display	Defective display, camera or combiner (if it's installed).	Connect the FOC from one camera directly to the display (if a combiner is used). If the problem still persists, replace camera and/ or display. Otherwise, replace the combiner.	n.a.
Bad image presentation	Misadjusted display parameters or defective display	Set Display into factory setting. If the problem still persists, replace the display.	n.a.

3.4 Status Information, Tests, PDD

Tab. 6 Information on troubleshooting PDD

Symptom	Explanation	Possible Cause	Service action
The Patient Data Display (PDD) shows the message 'Welcome to your MAGNETOM', but no image.	The Message 'Welcome to your MAGNETOM' is shown at the PDD when it receives no graphics data from the computer's graphics card. During system bootup, this is the normal state	The secondary port of the host's graphics interface is not activated or is configured incorrectly.	Check if the secondary output of the graphics card is activated. To do so, open Start-->Control-->Display-->Settings at the host computer. Make sure the screen resolution of the secondary display is activated and set to 1280 x 800 pixels. Set the position of the secondary display to (1280,-800), (➔ Fig. 16 Page 36). If two host displays are installed, the PDD is set to be the third display with position (2560,-800). If the display setting cannot be set to 1280 x 800 pixels, see “The screen resolution of the Patient Data Display (PDD) cannot be set to 1280 x 800 pixels”.
The screen resolution of the Patient Data Display (PDD) cannot be set to 1280 x 800 pixels	The graphics card in the host has not received valid information regarding which pixel settings are supported by the PDD. Since the system uses optical signal transmission, this information is stored in the DVI extender plug (also called "Stretch DVI").	The DVI extender was not installed during host bootup, the required pixel setting Information could not be read.	Make sure the DVI extender plug is connected to the host computer (an adapter cable or similar might be installed between the graphics card and DVI extender). Since there are separate transmitter and receiver plugs, make sure that the correct one is installed. Make sure all electrical connections are secured. Reboot the host computer.
		The DVI extender was not programmed during manufacturing.	Replace the DVI extender.
		The graphics port --> DVI adapter cable at the host is defective.	Replace the adapter cable (or adapter plug, if applicable).

Symptom	Explanation	Possible Cause	Service action
The Patient Data Display (PDD) screen does not display an image and is black	Missing supply voltage, missing graphics data, or an incorrect display position in the graphics card menu leads to a black screen.	No image information output, Syngo_MR_PDD_Server software is not operating properly.	Stop and restart the Syngo_MR_PDD_Server software at the host. Go to Start Menu --> Syngo --> Tool Box --> Component Viewer, search for Syngo_MR_PDD_Server (➔ Fig. 15 Page 35). Stop and restart the process using the menu available with the right mouse button.
		Incorrect position of secondary display in the graphics menu.	Make sure that the position of the secondary display is set to (1280,-800) in Start-->Control-->Display-->Settings (➔ Fig. 16 Page 36).
		No supply voltage at the PDD.	Make sure that the supply voltages are on. This is indicated by two small green LED lights located at the RFCEL satellite above connector plug X15 (for front display) or X16 (rear display). If the supply voltage is not on, replace the fuses. If this is not possible, replace the RFCEL satellite board.

Fig. 15: Component Viewer

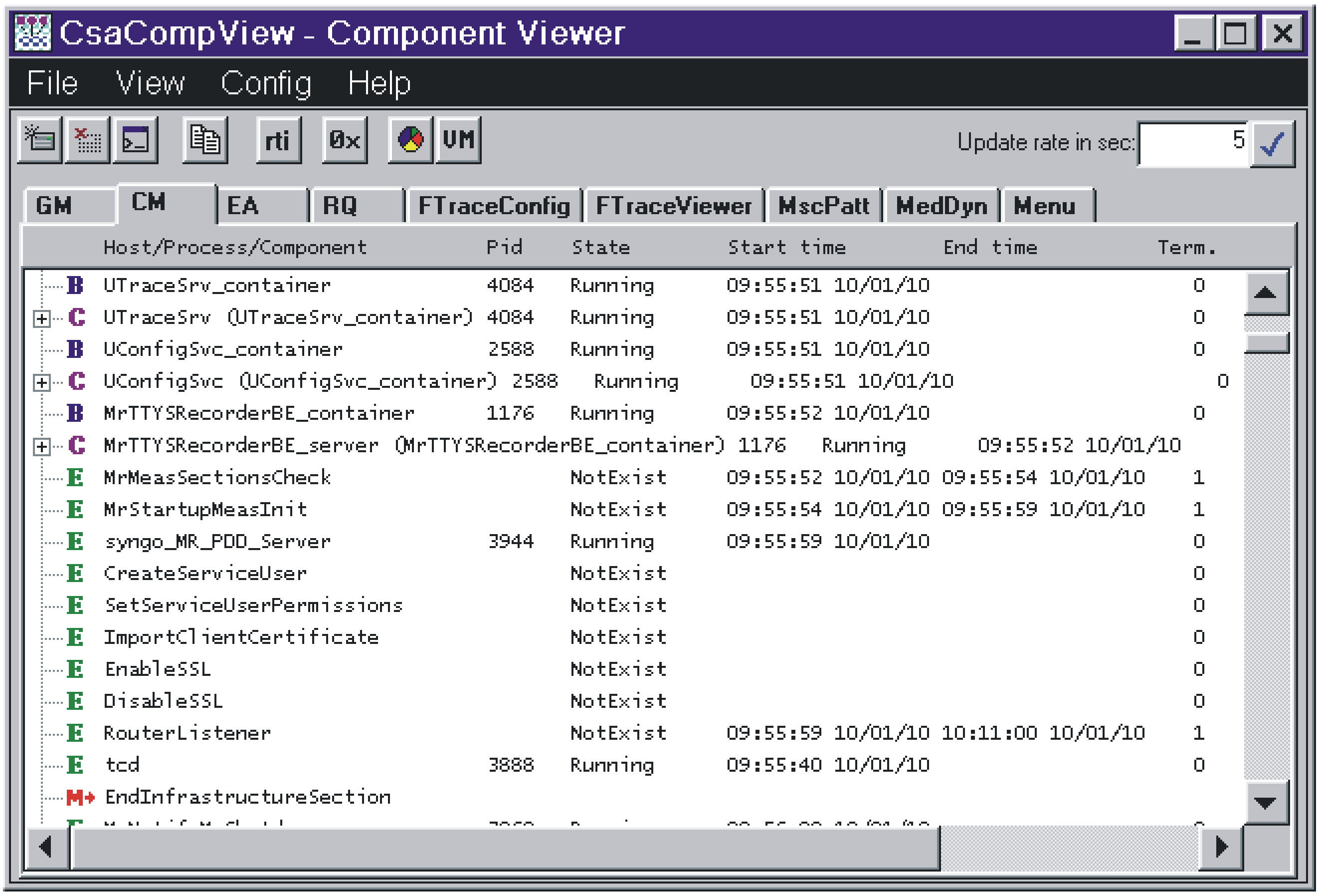
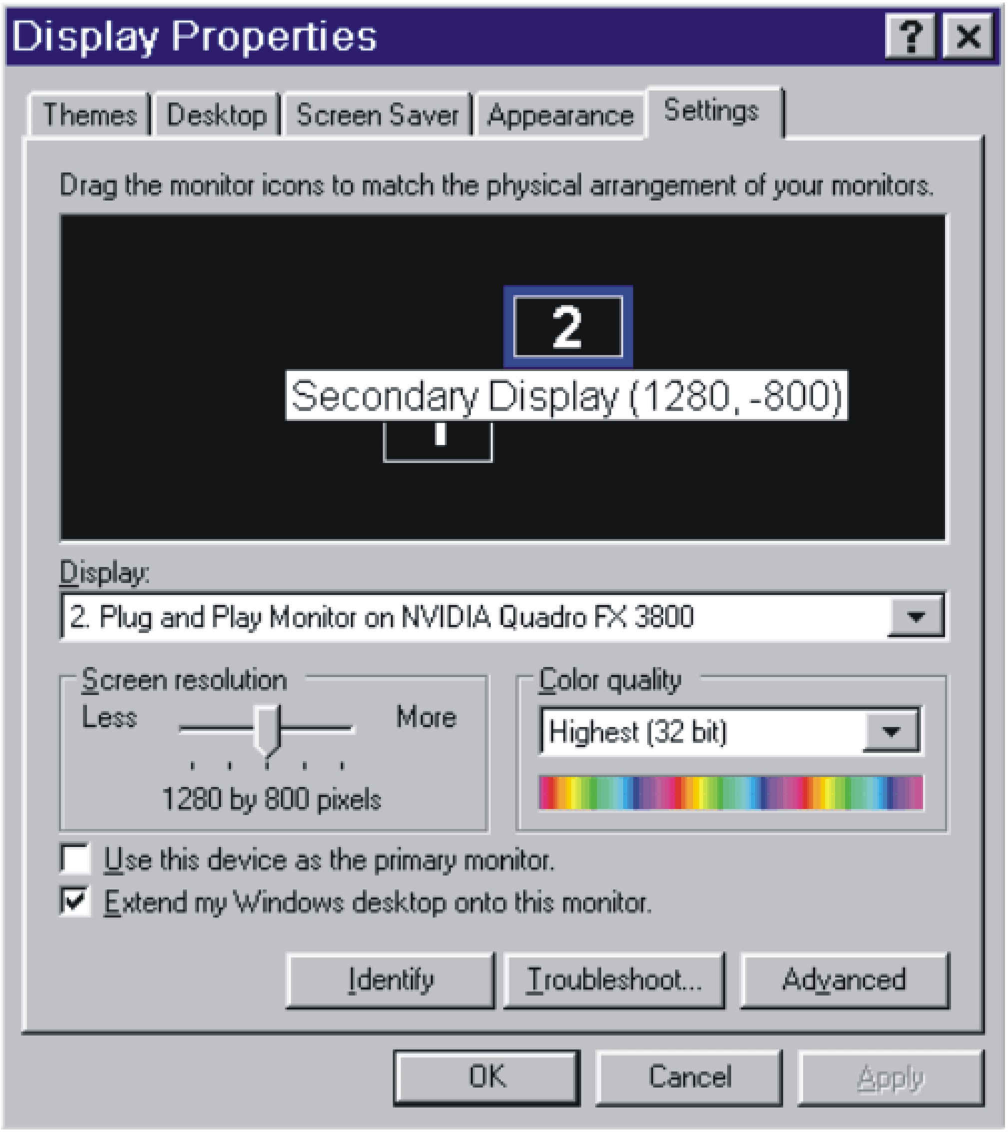


Fig. 16: Display Properties



3.5 Configuration

n.a.

4.1 Troubleshooting Strategy

This section of the TSG deals with problems regarding the Intercom. The strategies and procedures provide the necessary information for servicing the Intercom.

The primary components of the **Intercom** are:

- **Intercom operating unit**
- **Intercom board**- in the RFCEL
- **Loudspeaker** - in the magnet cover
- **Microphones** - in the magnet cover

The "Replacement of Parts" folder lists the necessary adjustments and the final checks when replacing the FRU.

4.2 LEDs, Test Points

n.a.

4.2.1 LEDs and Displays

n.a.

4.2.2 Test points

n.a.

4.2.3 Service switches

n.a.

4.2.4 Jumpers

n.a.

4.3 Status Information, Tests

Tab. 7 Informations on troubleshooting

Symptom	Probable Cause	Service action	Instructions
Patient can not be heard from the operating unit	**Magnet Cover** Microphones defective or not connected	Remove the covers front and rear side, check connection of the microphone amplifiers. Swap the front and the rear microphone amplifier board.	n.a.
	Operating Unit Loudspeaker or Audio Amplifier defective	Activate the ECG tone. If it is audible, the amplifier and loudspeaker are OK. If not, proceed with the following steps. If unsuccessful replace the operating unit.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Application Volume is set too low	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Check the volume from the operator to the patient and set it to maximum level, then reduce it to your desired level.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Patient can not heard operator from the headphone	**Patient Table** Sound transducer defective or not connected wire/tube	Remove the cover from the patient table (coil connector food end) and check the electrical connection. Check the pneumatical connection. Check, if the tube from the transducer is not bended.	n.a.
	Operating Unit Microphone or other components defective	Works the room speaker? If yes, the operating unit is not the problem. If not, proceed with the following steps.If no success exchange operating unit.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Application Volume is set too low	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Set the fader to center position. Check the volume from the patient to the operator and set it to maximum level, then reduce it to your desired level.	n.a.
	Wiring The signal path Intercom Board - Backplane - RCCS - wiring is broken	Remove the cover from the patient table (coil connector food end). Remove X135 from the RCCS and the ERNI Microspeed Pin for LC7 at the interconnection plate.Check the signal path: X135 D12 and X135 E12 to the separate clipped plugs at the ERNI.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Operator can not be heard from the loud-speaker	**Magnet Cover** Loudspeaker defective or not connected	Check if the music can be heard by the loudspeak-er. If that works, loudspeakers are o.k.	n.a.
	Operating Unit Microphone or other components defective defective	Check, if operator can be heard in the headphone. If that works, the operating unit is not the prob-lem.Works the room speaker? If not, proceed with the following steps. If no success exchange operat-ing unit	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, ben-ded areas etc. In such cases, the FOC has to be re-placed.	n.a.
	Application Volume is set too low	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Set the fader to center position. Check the volume from the operator to the patient and set it to maxi-mum level, then reduce it to your desired level.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output volt-age of the power supply at the removed cable , it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Music can not be heard from the headphone	**Patient Table** Sound transducer defective or not connected wire/tube	Remove the cover from the patient table (coil connector food end) and check the electrical connection. Check the pneumatical connection. Check, if the tube from the sound transducer is not bended.	n.a.
	Operating Unit Music in plug or other components defective	Works the room speaker? If yes, the operating unit is not the problem. If not, proceed with the following steps. If no success exchange operating unit	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Application Volume is set too low	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Check the volume for music and set it to maximum level, then reduce it to your desired level.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.
	Audio player Low battery, defective, wrong settings	Check the audio player with a separate headphone	n.a.

Symptom	Probable Cause	Service action	Instructions
Music can not be heard from the loudspeaker	**Magnet cover** Loudspeaker defective or not connected	Check if the music can be heard by the loudspeaker. If that works, loudspeakers are o.k. If not proceed with the following steps. If no success exchange the loudspeaker.	n.a.
	Operating Unit Music in plug or other components defective	Works the room headphone? If yes, the operating unit is not the problem. If not, proceed with the following steps. If no success exchange operating unit.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Application Volume is set too low	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Check the volume for music and set it to maximum level, then reduce it to your desired level.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.
	Audio player Low battery, defective, wrong settings	Check the audio player with a seperte headphone	n.a.

Symptom	Probable Cause	Service action	Instructions
Computer Voice can not be heard from the headphone	**Patient Table** Sound transducer defective or not connected wire/tube	Remove the cover from the patient table (coil connector food end) and check the electrical connection. Check the pneumatical connection. Check, if the tube from the transducer is bended.	n.a.
	Host / Application Wrong settings in Patient Comfort Configuration, Exam UI or Sound Card	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Set the fader to center position. Check the volume from the operator to the patient and set it to maximum level, then reduce it to your desired level. Then go to voice output and configure the playback volume in the Exam UI in the same way. This provides direct access to the sound card of the host. Managing the computer voice will be easier in later VD-XX versions.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Wiring The signal path Intercom Board - Backplane - RCCS - wiring is broken.The audio signal path from operating unit to the host is broken.	Remove the cover from the patient table (coil connector food end). Remove X135 from the RCCS and the ERNI Microspeed Pin for LC7 at the interconnection plate.Check the signal path: X135 D12 and X135 E12 to the separate clipped plugs at the ERNI. Check the audio cables from the connection box of the Operating Unit to the sound card of the host.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Computer Voice can not be heard from the loud-speaker	**Magnet cover** Audio amplifier defective, loudspeaker de-fective or not connected	Check if the music can be heard by the loudspeak-er. If that works, loudspeakers are o.k. If not pro-ceed with the following steps. If no success ex-change the loudspeaker	n.a.
	Host / Application Wrong settings in Patient Comfort Configu-ration, Exam UI or Sound Card	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Set the fader to center position. Check the volume from the operator to the patient and set it to maxi-mum level, then reduce it to your desired level. Then go to voice output and configure the playback volume in the Exam UI in the same way. This pro-vides direct access to the sound card of the host. Managing the computer voice will be easier in later VD-XX versions.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, ben-ded areas etc. In such cases, the FOC has to be re-placed.	n.a.
	Wiring The audio signal path from operating unit to the host is broken	Check the audio cables from the connection box of the Operating Unit to the sound card of the host.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output volt-age of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Computer Voice can not be heard from the operating unit	**Operating unit** Audio amplifier defective, loudspeaker defective or not connected	Activate the ECG tone. If this is audible, amplifier and loudspeaker are o.k. If not, proceed with the following steps. If no success exchange operating unit.	n.a.
	Host / Application Wrong settings in Patient Comfort Configuration, Exam UI or Sound Card.	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Set the fader to center position. Check the volume from the patient to the operator and set it to maximum level, then reduce it to your desired level.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Wiring The audio signal path from operating unit to the host is broken.	Check the audio cables from the connection box of the Operating Unit to the sound card of the host.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed.	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Computer Voice can not be recorded	**Operating Unit** Microphone defective or not connected.	Check, if the communication from operator to patient works. If yes, microphone is o.k.If not, proceed with the following steps. If no success exchange microphone.	n.a.
	Host / Application Wrong settings in Patient Comfort Configuration, Exam UI or Sound Card	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Set the fader to center position. Check the volume from the operator to the patient and set it to maximum level, then reduce it to your desired level. Then go to voice output and configure the record volume in the Exam UI in the same way. This provides direct access to the sound card of the host. Managing the computer voice will be easier in later VD-XX versions. You can only record your own computer voice sounds if a patient is not registered	n.a.
	Wiring The audio signal path from operating unit to the host is broken.	Check the audio cables from the connection box of the Operating Unit to the sound card of the host.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.

Symptom	Probable Cause	Service action	Instructions
Table Stop twinkling without activation from the Operating Unit	**Operating Unit** Supervision Circuit defective	Proceed with the following steps, if no success exchange the Operating Unit	n.a.
	Wiring Cable to patient table broken or disconnected, other any stop button pressed	Check the other stop buttons at the cover and the patient table, release them if activated. Remove cables and check the connection from PAM XB2--Operating unit connection box emergency stop cable W10161 X4 due to damages, bended areas etc. In such cases, the cable has to be replaced.	n.a.
	Patient Table Current Source defective.	Remove the cable from the power current from the connector box of the operating unit. Measure the output current of the source in the patient table PAM XB2, it must be 5 ..... 10mA with "+" line at the inside of the plug.	n.a.

Symptom	Probable Cause	Service action	Instructions
ECG not audible	**Operating unit** Audio amplifier defective, loudspeaker defective or not connected.	Check, if the communication from patient to operator works. If yes, Operating Unit is o.k. If not, proceed with the following steps. If no success exchange operating unit.	n.a.
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	n.a.
	Power The Operating Unit or the Intercom Board have no voltage feed.	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply at the removed cable, it must be 7.5V +/- 5%.	n.a.
	Host / Application Wrong settings in Patient Comfort Configuration, ECG button not activated.	**Open the Patient Comfort Configuration:** Options --> Configuration --> Patient Comfort. Check the volume for ECG and set it to maximum level, then reduce it to your desired level.	n.a.
	PMU SW on mars Not working, signal transmission from peru or ppu to RFCEL_COM (RF) broken, communication RFCEL_COM to MARS broken, communication from RFCEL_COM to Intercom Board broken	Check the PMU with tests belonging to it.	n.a.

Symptom	Probable Cause	Service action	Instructions
Stop sequence / table movement not possible from Operating Unit	**Operating unit** Button, relays, or electronics defective.	Proceed with the following steps, if no success exchange the Operating Unit.	n.a.
	Wiring Short circuit in the wiring from operating unit to the patient table.	Check if the STOP button at the Operating Unit twinkles even so the table movement works, if yes there must be a short circuit in the wiring PAM XB2--Operating unit connection box emergency stop cable W10161 X4.	n.a.
	Patient table Defective	Check the current supervision of the patient table.	n.a.
Reset stop sequence / table movement not possible from Operating Unit	**Operating unit** Button, relais or elctronic defective	Exchange the Operating Unit.	n.a.

Symptom	Probable Cause	Service action	Instructions
Patient Alarm can not be heard from operating unit	**Squeeze ball** Tube broken	Check the tube of the squeeze ball.	
	P/E transducer Tube not connected, cable not connected or transducer defective.	Remove the cover from the patient table (coil connector food end) and check the electrical connection. Check the pneumatical connection. Check, if the tube from the P/E-transducer is not bended.	
	FOC broken or not connected	Check the FOC on both ends due to damages, bended areas etc. In such cases, the FOC has to be replaced.	
	Wiring The signal path Intercom Board - Backplane - RCCS - wiring is broken.	Remove the cover from the patient table (coil connector food end). Remove X134 from the RCCS and the ERNI Microspeed Pin for LC8 at the interconnection plate. Check the signal path: X134 D12 and E12 to the separate clipped plugs at the ERNI.	
	Power the Operating Unit or the Intercom Board have no voltage feed	Check the supervision LEDs on the MSM of RFCEL for: 1.5V, P6.8V, P15V, P25V, N32V. Remove the cable from the power supply from the connector box of the operating unit. Measure the output voltage of the power supply, it must be 7.5V +/- 5%	
	Operating Unit Loudspeaker or Audio Amplifier defective	Activate the ECG tone or listen to the patient. If this is audible, amplifier and loudspeaker are o.k.	
Patient Alarm can not be reset from operating unit	**P/E transducer** defective, makes always contact without pressing squeezeball	Check the resistance at clipped plugs of ERNI Microspeed Pin for LC8. If there is a short circuit without pressing the squeeze bal, the P/E-transducer is defective.	

4.4 Configuration

n.a.

5.1 Troubleshooting Strategy

The following chapter describes the troubleshooting procedures for the **fixed** and the **mobile** patient table, of the MR Systems **Aera** / **Skyra** / **Prisma**

There are 2 versions of the patient table offered to the customer:

- **Tim Table (Fixed Table)**
- **Tim Dockable Table (mobile table)**

The main tool for patient table troubleshooting are the error messages delivered from the patient table. The error messages are shown in the error log file from the local service platform. A detailed description about the error is part of the error message. All message IDs are listed in the Healthcare Services Knowledge Base (SKB). Each error message includes the error ID, message text, explanation and action. The patient table communication and status test are also available in the Service Software platform under "TestTools" and in the Service Software for the patient table.

To supplement the error messages, additional tables provide the CSE with information on troubleshooting switches, sensors, jumpers, and LEDs. FRU replacements are described in the REP patient handling chapter (➔ FRU Overview / M7-000.841.15).

For an explanation of the basic functions of the patient table, refer to the Function Description in the patient handling chapter (➔ Interactive Functional Description / M7-000.850.01).

The "Replacement of Parts" folder lists the necessary adjustments and the final checks when replacing the FRU.

5.2 LEDs, Test Points, Fuses, Jumpers, Switches, Sensors

This chapter describe the LED´s, test points, fuses, jumpers, switches and sensors of the patient table.

5.2.1 LEDs and Displays

Tab. 8 LED’s

Location	Name	Description
PAM - D4120	H1	+5 V int.
PAM - D4120	H2	Temperature error
PAM - D4120	H3	+22 V int.
PAM - D4120	H4	+22 V out
PAM - D4120	H5	+60 V out
PAM - D4120	H6	+60 V int.
PSY - D4110	H10	+22 V int.

5.2.2 Test Points

Tab. 9 Test points

Location	Name	Description
PAM, cable W11160	XS1, Pin 1 PIN 4	+60 V GND
PAM, cable W11160	XS1, Pin 7 PIN 5	+22 V GND

5.2.3 Fuses

Tab. 10 Fuses

Location	Name	Description
PAM	F1	25 A, (spare fuse located on PAM)
PSY	F10	4 A, (spare fuse located on PSY)

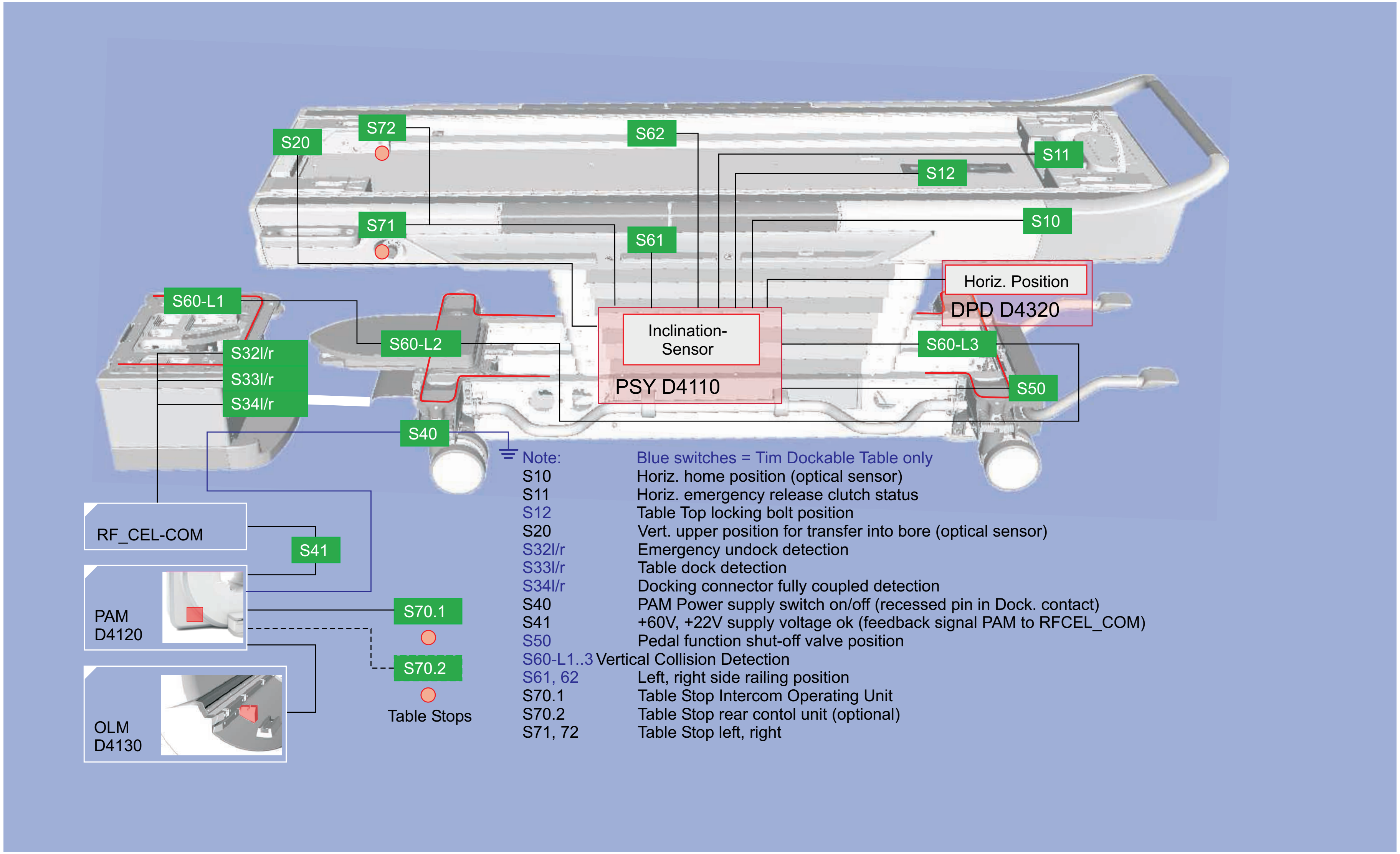
5.2.4 Jumpers

Tab. 11 Jumpers

Location	Name	Description
PAM - D4120	XB3	Must be removed with 3rd operation panel

5.2.5 Switches, Sensors

Fig. 17: Patient Table Switches and Sensors Overview



Tab. 12 Switches and functions

Location	Name	Description	Status
Foot-end, underneathSupport-Frame.	S10	Optical Sensor, Home reference Z-position.	Activated (=closed) in Home Position, vertical movement enabled.
Foot-end, underneath Support-Frame (mounted at emergency release mechanics).	S11	Emergency clutch for horizontal drive.	Switch is open when motor decoupled from traction chain (emergency clutch).
Foot-end top of Support Frame (underneath cover).	S12 (mobile only)	Table plate locking bolt, secures plate against unwanted horizontal movement.	Activated (=closed) when locking bolt is active (always when table is undocked and for a short time during S12 test, i.e. shortly before reaching the lowest vertical position).
Head-end of Support Frame.	S20	Optical sensor, detects vertical transfer position into magnet.	Activated (=closed) when reflector at magnet cover is reached..
Docking Station, at Docking hook.	S32l/r (mobile only)	Detection of emergency undock	Released (= open) when the table is emergency undocked.
Docking Station, at Docking hook.	S33l/r (mobile only)	PTAB in docking range, docking possible, or table is docked.	Actuated (= closed) when docking hooks are pushed to downward position.
Docking Station, at Docking Connector magnet-side	S34l/r (mobile only)	Docking connector fully coupled (sufficient contact pressure).	Actuated (= closed) when fully locked.
Docking Connector, X53	S40 (mobile only)	Recessed pin in Docking Connector (X53, pin-1). Detects PTAB docked to magnet.	Closed when Docking Connector fully coupled, (enables PTAB supply voltages 22 V and 60 V at PAM).
PAM	S41	Potential free voltage feedback PAM, XB7, pin-1+2 to RFCEL_COM.	Actuated (=closed) signal active (= low level) when 22 V and 60 V table supply from PAM is present.
At Y-Drive D4312, foot end.	S50 (mobile only)	Monitoring switch for hydraulic valve position, i.e. pedal function is disabled when Table Top is not in Home Position. Operated through hydraulic actuator from Table Top.	Actuated (= open) when Table Top is not in Home Position.

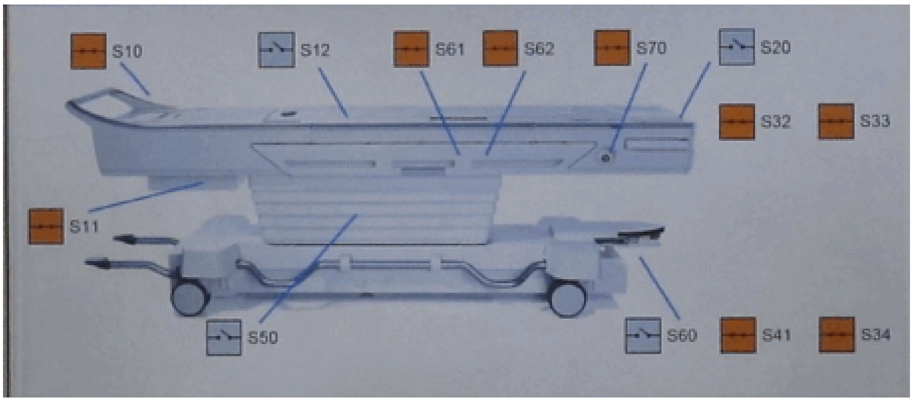
Location	Name	Description	Status
Top of PTAB chassis and Docking Station.	S60 L1, L2, L3 (mobile only)	3 switch rails for vertical collision detection (obstacle under table).	Actuated (= closed) when collision (a 1.2 kOhm terminator resistor connected in parallel for cable break detection).
Support frame, left-side.	S61 (mobile only)	Position of left side railing (horizontal movement enabled when hinged down).	Actuated (= closed) when left side railing is fully hinged down.
Support frame, right side.	S62 (mobile only)	Position of right side railing (horizontal movement enabled when hinged down).	Actuated (=closed) when right side railing is fully hinged down.
Stop Intercom Magnet cover rear	S70.1 S70.2	Stop Intercom PTAB Emergency Stop rear control unit, if optional rear panel is fitted, (connected in series to S71 and S72. If no S70.2 is fitted, a dummy plug must be installed at PAM connector XB3).	Actuated (= open) when intercom or rear table stop is pressed. !!With older SW versions the switch may be incorrectly displayed at the PDD as Actuated (= open) when the switch is NOT pressed!!
Support frame, left side.	S71	PTAB Emergency Stop 1	Actuated (= open) when table stop is pressed.
Support frame, right side.	S72	PTAB Emergency Stop 2	Actuated (= open) when table stop is pressed.

5.2.6 Status of PDD Patient Table in Home Position (Service Mode)

Log in to Local Service / TestTools / PTAB and compare the PDD switch status as shown in this picture.

Fig. 18: Patient table in home position

Fig. 19: PDD display, patient table in home position

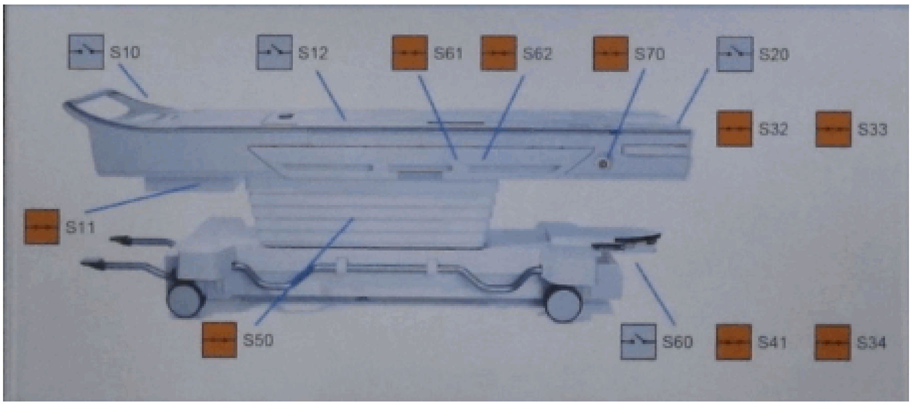


5.2.7 Status of PDD Patient Table in Scan Position (Service Mode)

Login to Local Service / TestTools / PTAB and compare the PDD switch status as shown in this picture.

Fig. 20: Patient table in scan position

Fig. 21: PDD display, patient table in scan position



5.3 Hints for Troubleshooting

Tab. 13 Informations on troubleshooting

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table is docked	**Horizontal drive**	Tabletop moves 5 mm in the direction of the isocenter and stops	Tabletop locking bolt did not not release the tabletop when table was docked	Check the mechanics of S12	(→ S12 Tabletop Locking Mechanics / Page 76)
		Tabletop does not reach correct home position (drags along magnet cover during vertical movement)	Tabletop extends beyond the support frame (on the side of the magnet)	Check the Home position (adjustment S10), tabletop must align to the support frame	(→ S10 Tabletop Home Position / Page 95)
Table is docked	**Horizontal drive**	Support frame drags along magnet cover during vertical drive	Table is located too close to the magnet	Check alignment of patient table at base plate, or check alignment of Docking Station.	see: **CB-DOC / Installation Instructions / Examination Room / Patient Table / Tim Dockable Table (Mobile Patient Table)** or **/ Fixed Patient Table (Tim Table)**

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table is docked	**Horizontal drive**	Large gap between support frame and magnet cover	Table is located too far off from magnet	Check alignment of patient table at base plate, or check alignment of Docking Station.	see: **CB-DOC / Installation Instructions / Examination Room / Patient Table / Tim Dockable Table (Mobile Patient Table)** or **/ Fixed Patient Table (Tim Table)**
		Emergency release handle of tabletop plate cannot be reset or pops out	Resetting fails	Pull the tabletop plate back and forth during the reset.	n.a.
			Emergency release is mechanically blocked	Check for obstacles. Check if cables in the vicinity of the lever are properly routed.	n.a.
Table is docked	**Horizontal drive**	Electrical horizontal movement is not possible	No operating voltage 22V or 60V is available.	Check if Table STOP is not activated. Start SeSo Test Tools and select the PTAB test to start the service menu at the PDD. Check switches S11, 20,61, 62. Check supply voltage PAM, check LED on PSY H10, check EventLog for PTDmsg.ID 114 “Under voltage” error	(➔ LEDs and Displays / Page 55)

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table is docked	**Horizontal drive**	Unexpected noise during horizontal movement, sporadically I^2 t error(over current)	Horizontal gear mechanical resistance too high.	Pull emergency release handle of tabletop plate, push table top into the bore and pull it back to Home Position, check for obstacles. The force of 150N should not be exceeded. Check clearance of tabletop plate by releasing the table top connection talon and moving the tabletop plate by hand. Check alignment of Tim Table at base plate, or check alignment of Docking Station.	(→ Removal / M7-000.841.15) and see: **CB-DOC / Installation Instructions / Examination Room / Patient Table / Tim Dockable Table (Mobile Patient Table)** or **/ Fixed Patient Table (Tim Table)**
Table is docked	**Vertical lift**	Error MRI_PTD 28 occurs	Table moves couple mm down during approx. 1 minute without any operator action.	Replace hydraulic cylinder vertical lift.	(→ Hydraulic cylinder - vertical lift / M7-000.841.15)
		Patient data disappear on the MRAWP after reaching the Home Position (VD13A downwards)			
		Electrical lift is not possible at all	No operating voltage 22V or 60V is available.	Check voltage supply PAM, check LED on PSY H10, check for EventLog PTDmsg.ID 79"Undervoltage" error	(→ LEDs and Displays / Page 55)
		Not possible to lower table electrically	Check vertical collision switches (S60 L1...L3) . When actuated, only up movement is possible	Check chassis and Docking Station covers	(→ Checking the Terminal Strip S60 (Vertical Collision Detection) / Page 80)
			Quick coupling is defective	Check hydraulic quick couplings	(→ Checking Hydraulic Quick Couplings / Page 82)

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table is docked	**Vertical lift**	Did not find correct height to enter magnet	Reflector is contaminated	Clean reflector	n.a.
			Optical height sensor S20 is defective	Check sensor (red light dot must be visible on the cover). Start SeSo / TestTools and select the PTAB test to start the service menu at the PDD. Drive table from min. to max vertical height and check the S20 switch activity at the PDD. If sensor aligns with the reflector, S20 must be closed. If not, check cable connections or replace S20	(➔ Handle head side, S10, S20, S61 (S62), S71 (S72) / M7-000.841.15)
			Opening hole at S20 optical height sensor is obstructed or covered	Clean opening hole at S20	n.a.
		Vertical table top entrance height is too low / high	Reflector is not adjusted properly	Clip out reflector from the magnet cover and adjust the bracket behind	(➔ Table movement / M7-010.831)
			Reflector is contaminated	Clean reflector	n.a
Table is docked	**Vertical lift**	Vertical table top entrance height is too low / high	Table is completely mechanically misadjusted. Optical height sensor S20 is not aligned to the reflector at magnet cover	Check alignment of Tim Table or Docking Station.	see: **CB-DOC / Installation Instructions / Examination Room / Patient Table / Tim Dockable Table (Mobile Patient Table)** or **/ Fixed Patient Table (Tim Table)**

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table is docked	**Emergency undocking**	Emergency undocking handle does not work. Handle can be pulled out without resistance	Cable pull is damaged	Check mechanics for damage	n.a.
		Emergency undocking handle mechanical resistance too high or blocked	Mechanics is blocked	Check/lubricate guides handle, guides and cable pull, use suitable lubricant, e.g. silicone spray, check routing at cable pull	n.a.
Table undocked	**Tabletop locking mechanics**	Tabletop plate can be horizontally moved by hand	Tabletop locking bolt is not functional.	Table must **be docked** or **simulated as docked**: Check S12 table top locking mechanics. Check S12 switch test assembly by moving the table fully down while monitoring for error messages at the PDD	(➔ S12 Tabletop Locking Mechanics / Page 76)

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table un-docked	**Vertical lift with ped-als not possible**	Retractable cover of the table re-mains open (pedals can be pressed to-wards the floor with increased force only)	Retractable cover is jammed in guiding	Check mechanics for damage, lubricate with silicone spray Check tension of bowden cable 5th wheel	(→ Checking for Damage of Bowden Cables / Page 100)
			Bowden cable is damaged	Visual inspection	(→ Checking for Damage of Bowden Cables / Page 100)
			Hydraulic connections loosen	Check hydraulic connections	n.a
		Pedals can be pressed towards the floor without resistance	Retractable cover jammed in guide	Check mechanics for damage, lubricate with silicone spray	(→ Checking the Docking Sta-tion Retractable Cover Mecha-nism / Page 104)
			Lift-Docking function select mechanics faulty	Check Lift-Docking function se-lect mechanics valve adjust-ment	(→ Lift-Docking Switch-Over / Page 87)
			Bowden cable is damaged	Visual inspection	(→ Checking for Damage of Bowden Cables / Page 100)
		Pedals can be pressed towards the floor with too high resistance	Hydraulic lock function in-complete	Check hydraulic lock adjust-ment	(→ S50 Hydraulic Lock Adjust-ments / Page 84)
	Moving the table	Table gets stuck e.g. at door thresh-old	5th wheel is damaged	Check mechanics for damage	(→ Checking 5th the Wheel / Page 94)
		Lateral table move-ment is difficult	5th wheel mechanics is blocked	The 5th wheel is engaged (op-erator fault). Disengage 5th wheel Check/adjust 5th wheel locking mechanics	(→ Checking 5th the Wheel / Page 94)

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table un-docked	**Moving the table**	Straight ahead table movement is difficult	5th wheel locking mechanics malfunction	The 5th wheel is disengaged (operator error). Engage 5th wheel. Check/adjust 5th wheel locking mechanics	(➔ Checking 5th the Wheel / Page 94)
			Gas pressurized spring is defective/wheel has no contact with the floor	Check the gas pressurized spring	(➔ Checking 5th the Wheel / Page 94)
		Retractable cover at Docking Station remains open after undocking procedure	Retractable cover is jammed in guiding, or mechanical resistance of actuator too high	Check mechanics for damages/ lubricate with silicone spray	(➔ Checking the Docking Station Retractable Cover Mechanism / Page 104)
			Object blocks retractable cover	Remove object	n.a.
To dock	**Table cannot be docked**	Pedals can be pressed toward the floor without resistance	Hydraulic lock function incomplete	Check hydraulic lock adjustment	(➔ S50 Hydraulic Lock Adjustments / Page 84)
			Lift-Docking function select mechanics is stuck in intermediate position, final position cannot be reached	Check Lift-Docking function select mechanics valve adjustment	(➔ Lift-Docking Switch-Over / Page 87)

State	Function	Symptom	Probable Cause	Service Action	Instructions
To dock	**Table cannot be docked**	Pedals can be pressed toward the floor with too high resistance	Hydraulic lock function incomplete	Check hydraulic lock adjustment	(➔ S50 Hydraulic Lock Adjustments / Page 84)
			Docking mechanics, mechanical resistance too high or low	Check docking mechanics for excessive resistance Check mechanical alignment	see: **CB-DOC / Installation Instructions / Examination Room / Patient Table / Tim Dockable Table (Mobile Patient Table)** or **/ Fixed Patient Table (Tim Table)**
			Lift-Docking function select mechanics switches incomplete	Check mechanic adjustment	(➔ Lift-Docking Switch-Over / Page 87)
		Retractable cover at Docking Station remains open after undocking procedure	Retractable cover is jammed in guiding, or mechanical resistance of actuator too high	Check mechanics for damages/ lubricate with silicone spray	(➔ Checking the Docking Station Retractable Cover Mechanism / Page 104)
		Error message "Emergency undock has been pulled"	Emergency undock has been pulled	While table in Docking Station, press undocking pedal fully toward floor, afterward docking is possible again.	n.a.

State	Function	Symptom	Probable Cause	Service Action	Instructions
To dock	**Table cannot be docked**	Table cannot be inserted into Docking Station	Alignment of table is not correct	Check the table alignment	see: **CB-DOC / Installation Instructions / Examination Room / Patient Table / Tim Dockable Table (Mobile Patient Table)** or **/ Fixed Patient Table (Tim Table)**
			Retractable cover of Docking Station or table is blocked/ does not open	Check the mechanics of the retractable cover of Docking Station or table	(→ Checking the Docking Station Retractable Cover Mechanism / Page 104)
			Docking Station locking hook is mechanically blocked	Press hooks down manually and check swivel range	n.a.
			Obstacle near docking connectors	Remove obstacle/objects	n.a.
			Bent contact at docking connectors	Replace docking connectors	(→ ODU connectors, mobile table, docking station / M7-000.841.15)

State	Function	Symptom	Probable Cause	Service Action	Instructions
Undocking	**Table cannot be undocked**	Pressing the undock pedal without effect. Pedal resistance is noticeably high	Table wheel brake engaged	Release brake	n.a.
			Tabletop plate is not fully in Home Position	Pull emergency release of tabletop plate, move table top into Home Position, press undock pedal. If pedal undock function is not successful, pull emergency undocking handle	n.a.
			Hydraulic lock function incomplete	Check hydraulic lock adjustment	(➔ S50 Hydraulic Lock Adjustments / Page 84)
		Table undocks partially cannot be undocked fully	Table locking hook is mechanically blocked	Push undock pedal twice, if not successful pull emergency release undock handle.	n.a.
		Pressing the undock pedal without effect. Pedal resistance is noticeably low	Hydraulic lock function incomplete	Check hydraulic lock adjustment	(➔ S50 Hydraulic Lock Adjustments / Page 84)

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table Stop or Squeeze-ball does not work properly	**Table Stop or Squeeze ball**	Table Stop on, left, right, rear, Inter-com, or squeeze ball does not work properly	Defective or disconnected FOC´s, hardware problem (PAM, PSY, Intercom board, cables, squeeze ball)	Disconnect FOC U1/2 at Inter-com and switch Supervision Mask for Intercom into state “Off” and “Virtual”. » Squeeze ball and all In-tercom parts are disa-bled.	If Scanner is **ok**: ■ Connect FOC U1/2 again and switch supervision on and real. ■ Make a wire bridge at the transducer connector X104 and reboot Scanner. » **If not ok** ■ Check FOC at Intercom board and Intercom itself, **or** take the service plug and connect it to the ERNI X134 at RCCS and press the test button on the service plug » **If ok** ■ Check squeeze ball.

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table Stop or Squeeze-ball does not work properly	**Table Stop or Squeeze ball**	Table Stop on, left, right, rear, Inter-com or squeeze ball does not work properly	Defective or disconnected FOC´s, hardware problem (PAM, PSY, Intercom board, cables, squeeze ball)	Disconnect FOC U1/2 at Inter-com and switch Supervision Mask for Intercom into state "Off" and "Virtual". » Squeeze ball and all In-tercom parts are disa-bled.	If Scanner **not ok**: ▪ Remove plugs XB2 and XB3 from the PAM and make a wire bridge directly at PAM connector XB2 pins 1 +6 and XB3 pins 1 +6. » **If ok** ▪ Connect X3 and try again » **If ok** ▪ check cabling X3 and Stop rear button. Otherwise, continue with next step. ▪ Connect X2 and disconnect X4 at the Intercom Inter-face I/O at the console. Es-tablish a wire bridge at the cable which was connected at X4 » **If ok** ▪ get new Operating Unit in-cluding the Interface I/O

State	Function	Symptom	Probable Cause	Service Action	Instructions
Table Stop or Squeeze-ball does not work properly	**Table Stop or Squeeze ball**	Table Stop on, left, right, rear, Inter-com or squeeze ball does not work properly	Defective or disconnected FOC´s, hardware problem (PAM, PSY, Intercom board, cables, squeeze ball)	Disconnect FOC U1/2 at Inter-com and switch Supervision Mask for Intercom into state “Off” and “Virtual”. » Squeeze ball and all In-tercom parts are disa-bled.	» If **not ok** ▪ make the wire bridge at the RF-filter plate connection X33A » If **not ok** ▪ Disconnect cable X4 at the PAM measure into the ca-ble W14204/14304 X4 pin 9 and 10 ▪ You have to measure a short circuit ▪ In case of an open connec-tion, proceed with the test at the connection X53 pins 4 and 5 Docking Station and X10 pins 9 and 10 PSY. Must be a short each time

5.3.1 Procedures (Mechanical Adjustments)

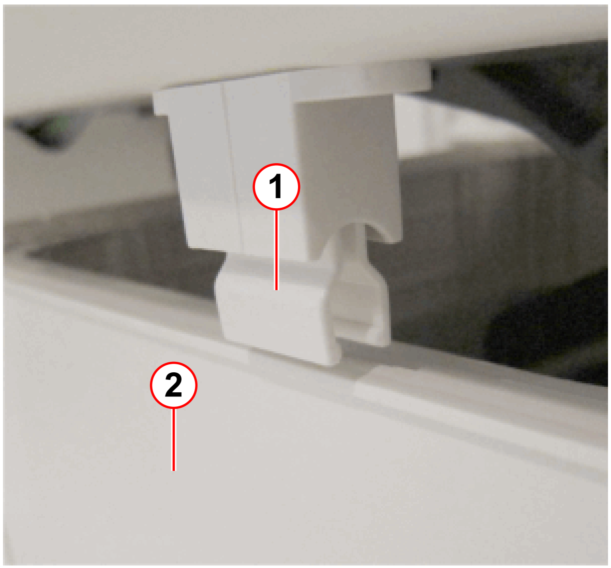
There are 2 types of plastic brackets used for the retractable cover.

Older version, 1 hole, until serial number 1028 for 3T and 1046 for 1,5T.

Latest version, 2 holes, form serial number 1029 for 3T and 1047 for 1,5T.

ONLY for Fixed Table. When the column cover is placed on the lower table chassis, do not drive the fixed patient table into the lowest position, the clips may collide and cause damage.

Fig. 22: Clips in collision with column cover



(1) Clip
(2) Column cover

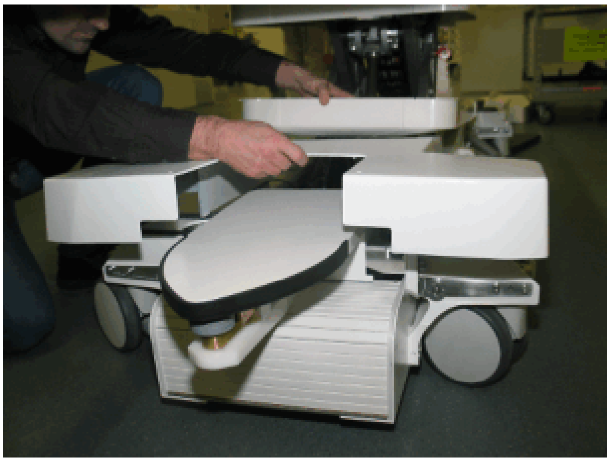
When driving the Fixed Table to a low table position, the clip (1) can collide with the column cover which is placed on the lower table chassis.

5.3.1.1 S12 Tabletop Locking Mechanics

Table is undocked and carriage cover has been removed

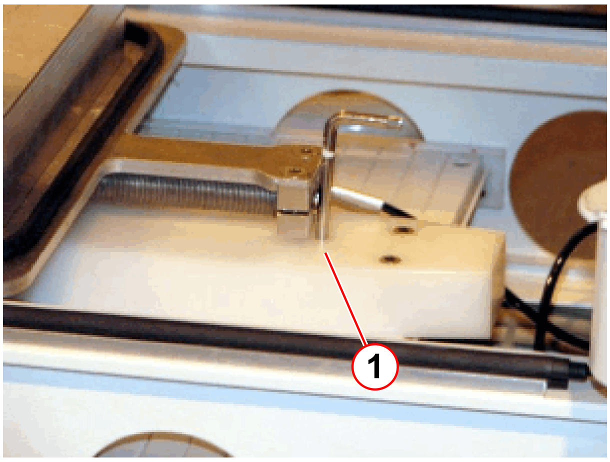
- Undock the Tim Dockable Table and move the table out of the magnet room.

Fig. 23: Chassis cover head end



- Remove the screws of the cover chassis head end.

Fig. 24: Retractable cover blocked

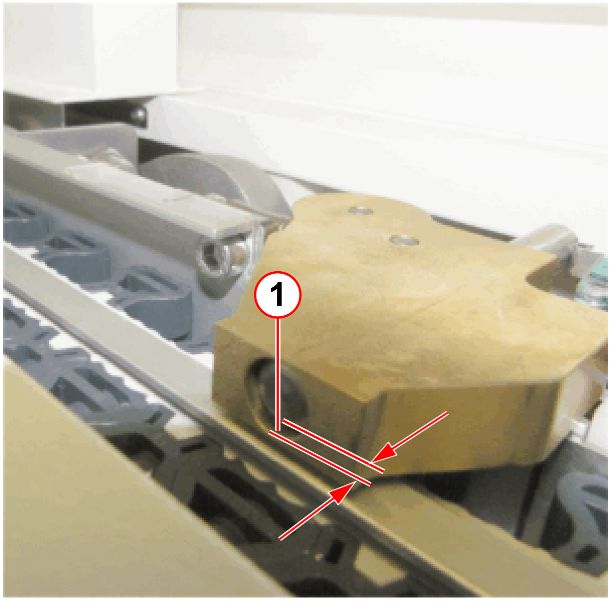


(1) 5mm allen key

- Pull the retractable cover back into the table.
- Insert a **5-mm** Allen key into the hole of the plastic bracket from the retractable cover and block the cov- er from closing. **The retractable cover must remain in the position as shown.**
  - » There are 2 versions of plastic brackets mounted (1 hole older version, 2 holes latest version)
  - » With the latest version use the second hole from the right side of the bracket

- Push the tabletop plate approx. 1000 mm away from Home Position.

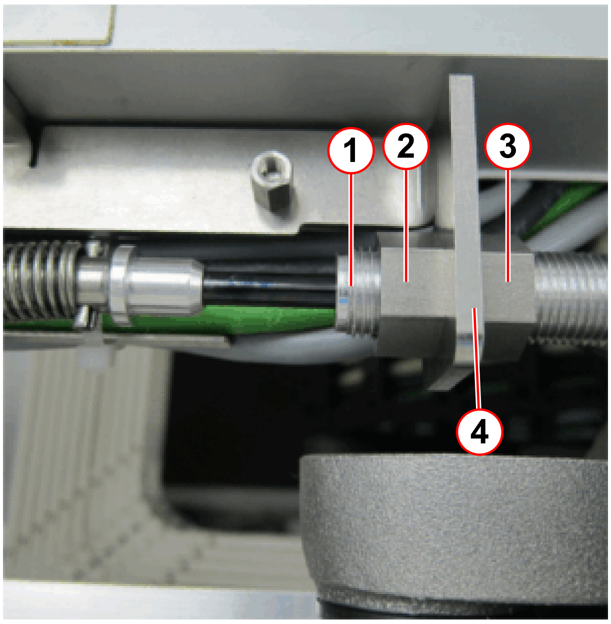
Fig. 25: Adjustment check S12



(1) Gear rack approx. 0-1mm inside the S12 housing

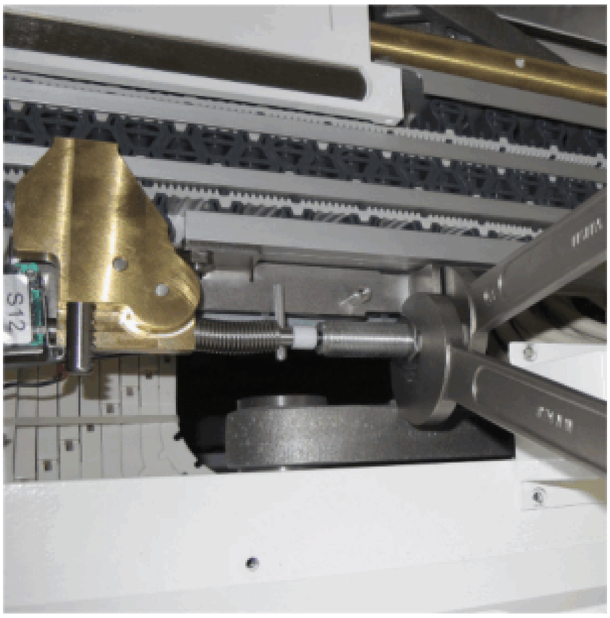
- Check the distance of the gear rack, the gear rack must be approx. 0-1 mm inside the S12 housing, as shown by the arrows (➔ 1/Fig. 25 Page 77).
- Adjust the 0-1 mm distance of the gear rack with the hex nut and counter nut at the hydraulic actuator by tightening the hex nut (2) by hand. Stop pushing the gear rack.

Fig. 26: Mounting the hydraulic actuator



(1) Hydraulic actuator
(2) Hex nut
(3) Counter nut
(4) Bracket

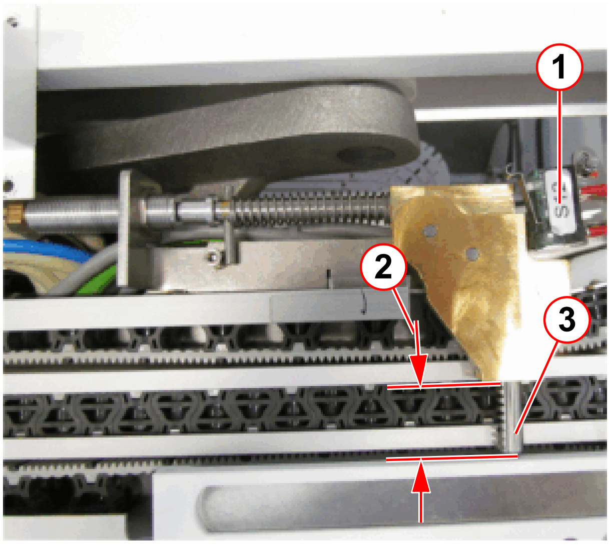
Fig. 27: Tightening the hydraulic actuator



- Tighten the hex nut with a 19-mm open wrench; use a second 19-mm open wrench, delivered with the update kit, to hold the counter nut.

  Double-check that the distance of 0-1 mm did not change.

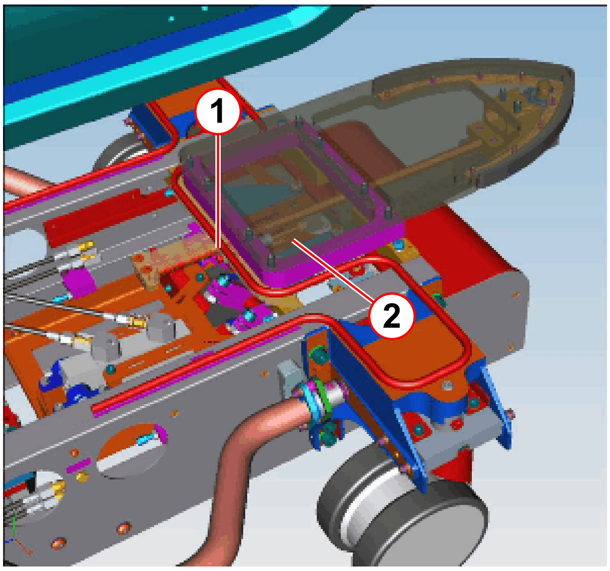
Fig. 28: S12 adjustment check with undocked patient table



(1) Electrical switch S12
(2) Minimum distance 20 mm from gear rack end to S12 housing
(3) Gear rack

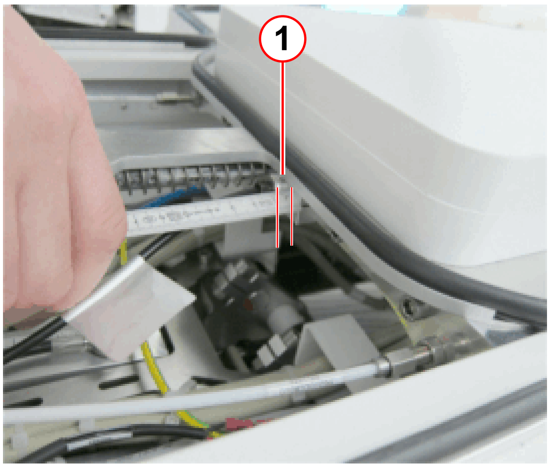
- The distance between the gear rack and the S12 housing must be **minimum 20 mm**.

Fig. 29: S12 location/adjustment



(1) Actuator thread, adjusted to min. of 2 mm
(2) S12 location

Fig. 30: Distance check/adjustment of the hydraulic actuator head-end chassis.



(1) Minimum length is 2 mm of the visible thread

- Only in the case that the 20 mm minimum distance is not reached, check the distance of the hydraulic actuator positioned at the bottom head-end chassis. The distance of the visible hydraulic actuator thread (1) at the mounting block outside can be adjusted to a **minimum of 2 mm**. Minimizing this distance increases the 20 mm distance of the S12 gear rack.

Note: Only in case the 20 mm minimum distance of the gear rack cannot be reached, replace the hydraulic actuator (not part of the update kit).

5.3.1.2 Final Check

- **Open the retractable cover and secure it.**
- Bring the tabletop plate into the Home Position and push the release handle inside.
- **Close the retractable cover.**
- Push the tabletop plate forward into the mechanical limit (approx. 6 mm) and dock the table - table docking must be possible.
- Switch on the system.
- Drive the tabletop plate electrically approx. 1 m inside the magnet.
- Check the 0-1 mm distance of the S12 gear rack again.
- Drive the tabletop plate electrically into the Home Position.

- Close all covers. For an easier installation of the column covers, we recommend to drive the patient table into the lowest position.
- Drive the table 5 cycles in and out and 2 cycles down and up, check for a smooth vertical movement of the table.
- Check the error messages in the system EventLog.

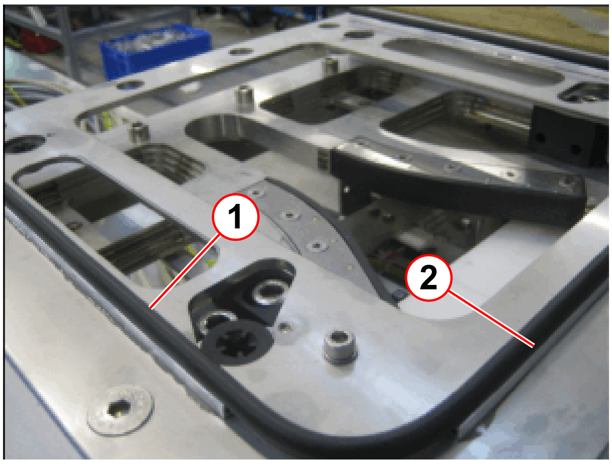
  If switch S12 is incorrectly adjusted, the following error messages are shown:
  - "MRI_PTD ID 19 switch error 3"
    and / or
  - "MRI_PTD ID 20 switch error 4"

5.3.1.3 Checking the Terminal Strip S60 (Vertical Collision Detection)

Docking Station

- Remove the Docking Station cover and check for damage.

Fig. 31: Docking station



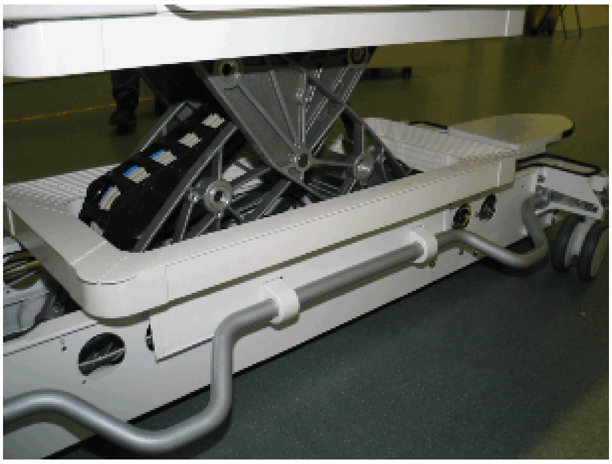
(1) Terminal strip S60
(2) Recess

- Check the course of the terminal strip (1). Ensure, that it is seated properly in the recess (2).
- If the error continues, replace the terminal strip.

Chassis

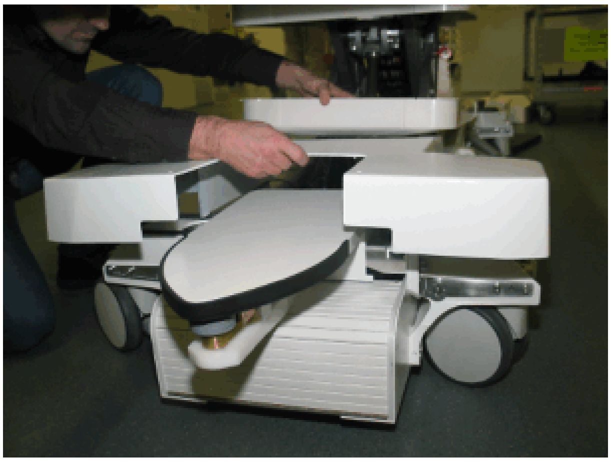
- Remove covers

Fig. 32: Table column cover



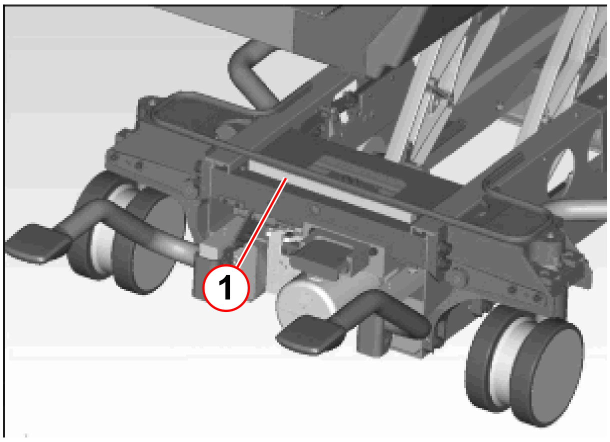
- Remove the table column cover.
- Insert safety bolts.

Fig. 33: Chassis cover head end



- Remove the screws of the cover chassis head end.
- Check the chassis covers for damage.

Fig. 34: Chassis



(1) Terminal strips

- Remove the chassis cover foot end.
- Check the course of the terminal strips (1). Ensure, that they are seated properly in the recess.
- If the error continues, replace the terminal strip.

- Remove safety bolts.

- Install covers.

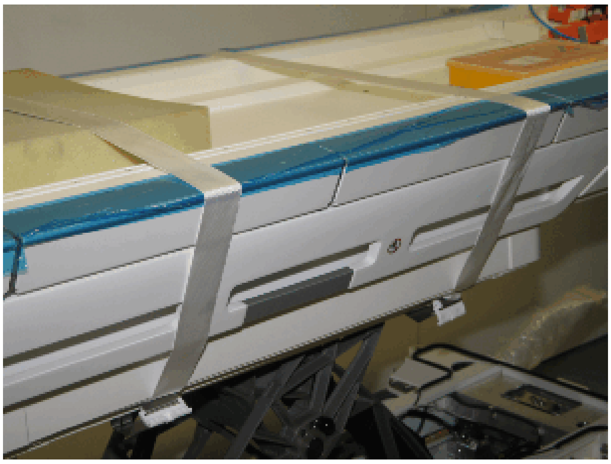
5.3.1.4 Checking Hydraulic Quick Couplings

Hydraulic-Pump Assembly, Cylinder, Dock mechanics

- Undock the Tim Dockable Table and move the table out of the magnet room.

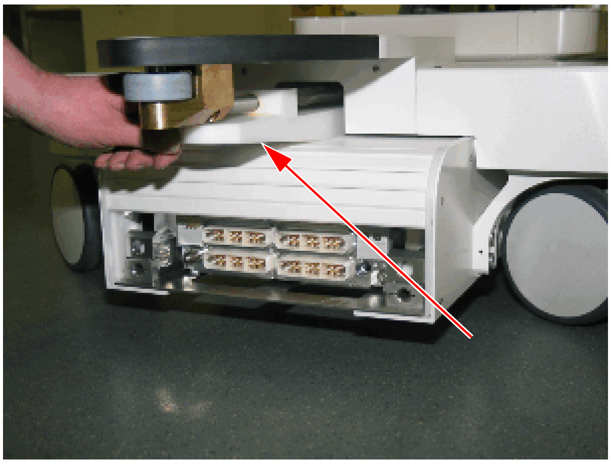
> The hydraulic system must without pressure for checking hydraulic quick couplings.

Fig. 35: Table column cover removed and secured



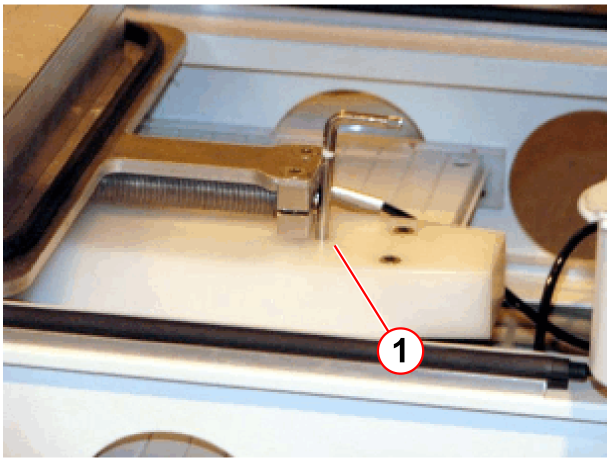
- Remove the table column cover.
- Insert safety bolts.
- Remove chassis cover or base frame covers.

Fig. 36: Actuator of the retractable cover, mobile table



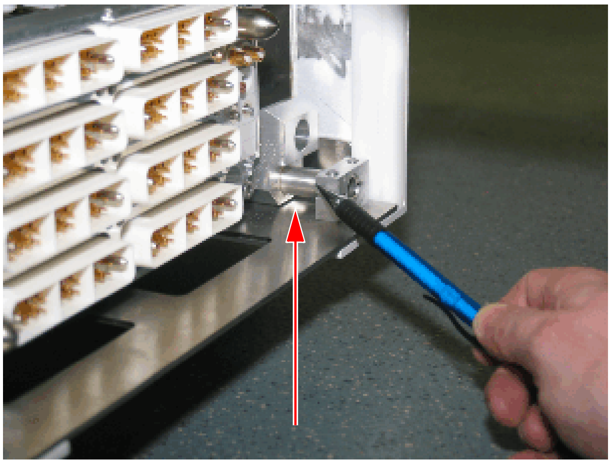
- Push the actuator of the retractable cover back by hand. Remove screws and lift-off the foot socket cover
- Secure the retractable cover against closing by inserting a 5-mm Allen key into the hole as shown in the picture on the left-hand side.
  - » Ensure, that the retractable cover is blocked.

Fig. 37: Actuator of the retractable cover blocked



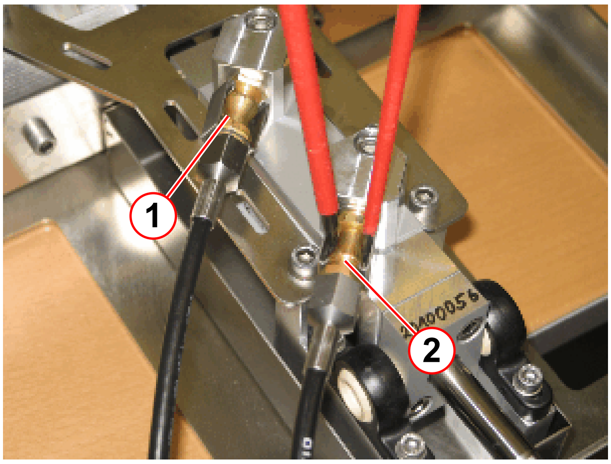
(1) 5mm allen key

Fig. 38: Hydraulic pusher



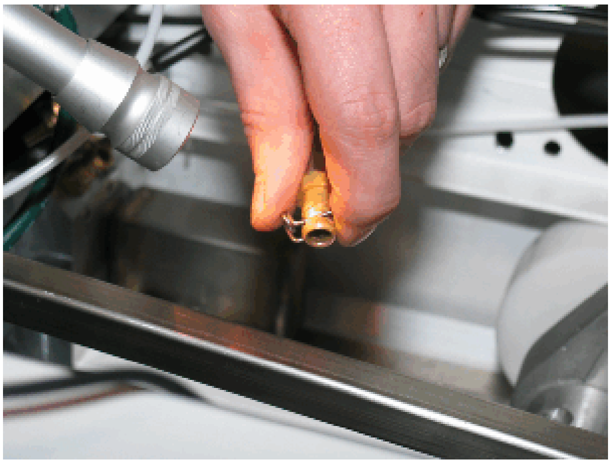
- Push the right food pedal a little until the hydraulic pusher is visible as shown in the left-hand picture.
  - » The hydraulic system is now without pressure.

Fig. 39: Hydraulic quick couplings



(1) Hydraulic quick coupling (new version)
(2) Flat blade screw drivers (for opening)

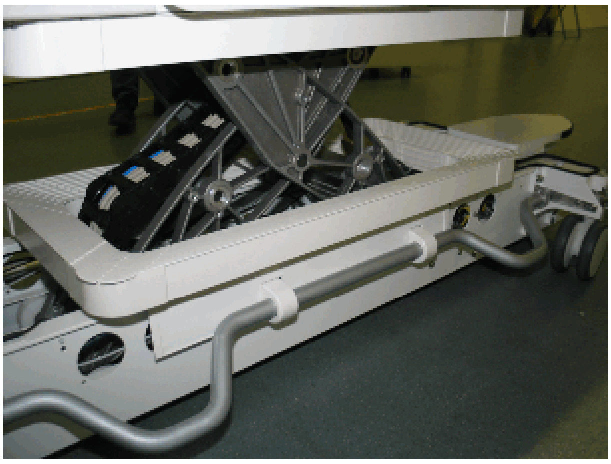
Fig. 40: Hydraulic valve



- Disconnect the quick couplings in the affected supply lines.
- Disconnect the lifting drive quick couplings, by using 2 flat blade screwdrivers (2).
- It should be possible to push in the quick coupling plunger slightly using a small screwdriver.
- Remove the hydraulic connectors by pushing the clamp downward and pulling the hydraulic hose back at the same time.
- It should be possible to push in the quick coupling plunger slightly using a small screwdriver.

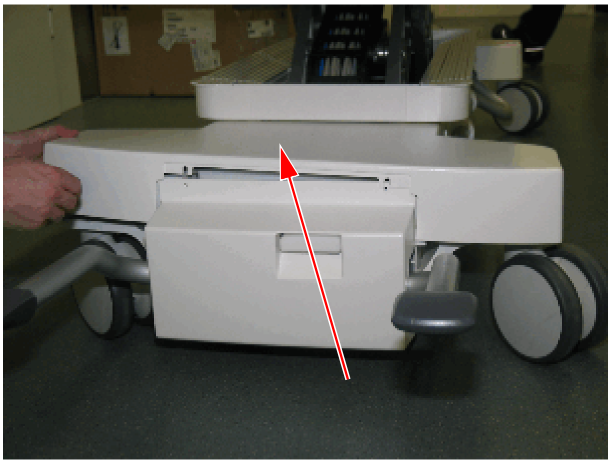
5.3.1.5 S50 Hydraulic Lock Adjustments

Fig. 41: Table column cover



- Remove the table column cover.

Fig. 42: Cover chassis - foot end top



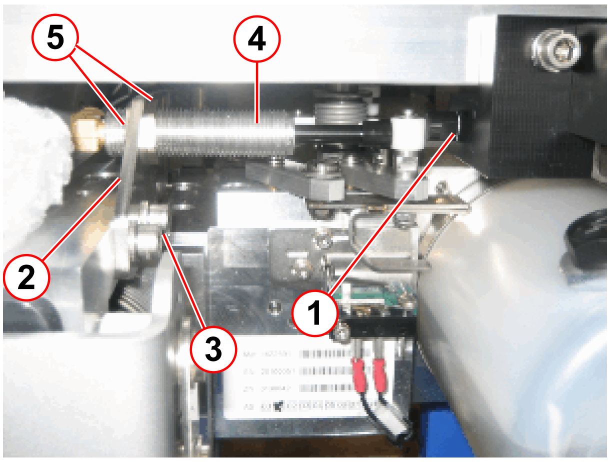
- Remove the cover chassis of the foot end top.

Table is docked or undocked

- For an exact position, drive the tabletop plate **electrically into the Home Position**.

> i The adjustment can be also performed if the table is undocked and the tabletop plate is in Home Position.

Fig. 43: Hydraulic actuator of lock function (S50)



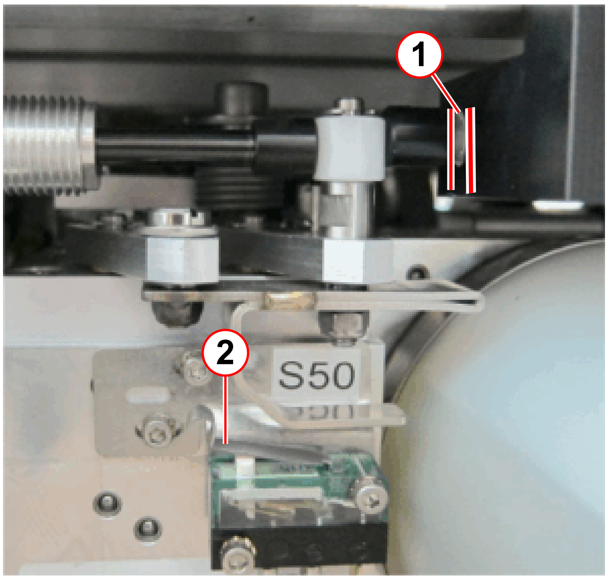
(1) Distance is between 1-2 mm
(2) Bracket
(3) Allen screws
(4) Hydraulic actuator
(5) Nuts to fix/adjust hydraulic actuator

- Check the distance (1). It must be **between 1-2 mm**, refer to details in (➔ 1/Fig. 44 Page 86).
- Skip the next 3 work steps if the measured 1-2 mm distance is correct.

The following 3 work steps are only needed when the adjustment requires correction:

- Remove the 2 Allen screws (3) of the bracket (2).
- Release the left and right nut (5). Move the hydraulic actuator (4) to the right until the **1-2 mm** distance (1) is set.
- Re-tighten the new hex nuts of the hydraulic actuator and tighten the Allen screws (3) of the bracket (2).

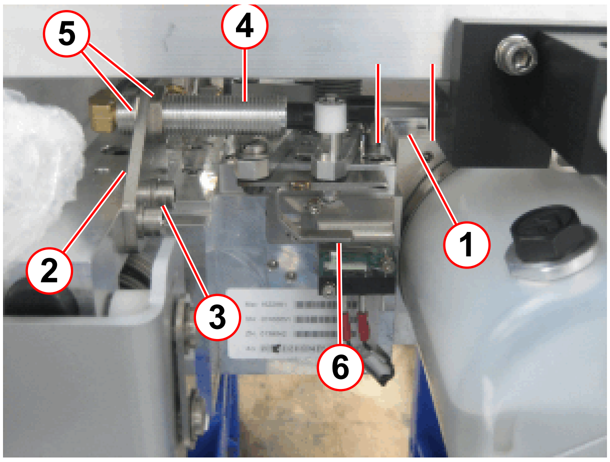
Fig. 44: S50 distance 1-2 mm check/adjustment



(1) Distance to be checked/adjusted to 1-2 mm
(2) Electrical switch S50 is deactivated

- Distance (1) must be **between 1-2 mm**.
- Electrical switch S50 (2) must be de-activated.

Fig. 45: S50 overview

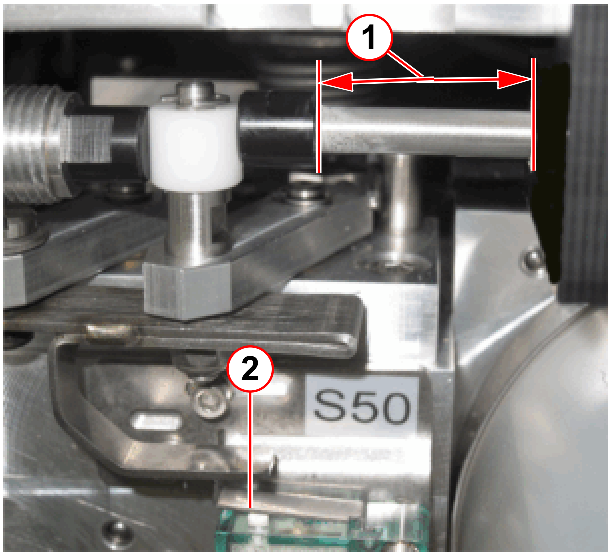


(1) Distance is min.16- mm
(2) Bracket
(3) Allen screws
(4) Hydraulic actuator
(5) Nuts to fix/adjust hydraulic actuator
(6) Switch plate

- Drive the tabletop plate electrically by approx. 500 mm toward magnet direction.
- Check the distance (1) again, the value must be a minimum of 16 mm, refer to details in (→ 1/Fig. 46 Page 87)

Note: If the min. 16-mm distance cannot be reached, replace the hydraulic actuator (4) (not included in the update kit).

Fig. 46: Checking of S50 distance - minimum 16 mm



(1) Distance of minimum 16 mm to be checked
(2) Electrical switch S50 is activated

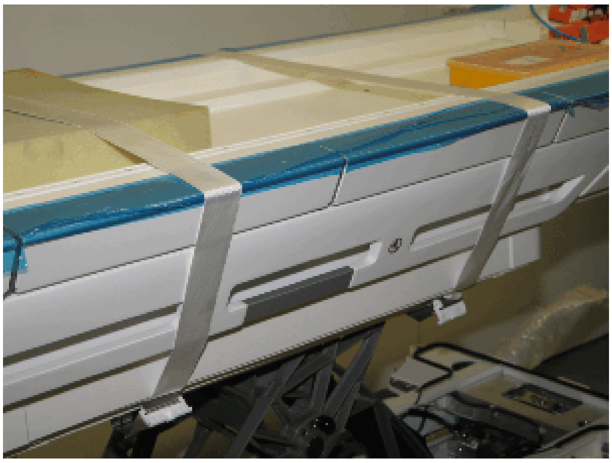
- Distance (1) must be **minimum 16 mm.**.
- The electrical switch S50 (2) must be activated.

- Install all covers.
- Drive the tabletop plate into the Home Position.

5.3.1.6 Lift-Docking Switch-Over

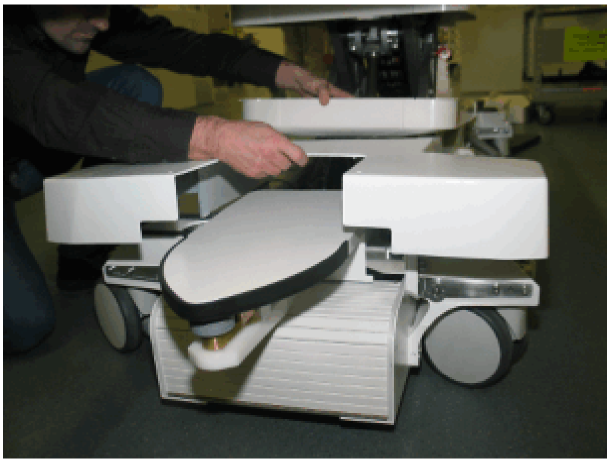
- Undock the Tim Dockable Table and move the table out of the magnet room.

Fig. 47: Table column cover removed and secured



- Remove the table column cover.
- Insert safety bolts.

Fig. 48: Chassis cover head end



- Remove the screws of the cover chassis head-end.

Fig. 49: Table column cover

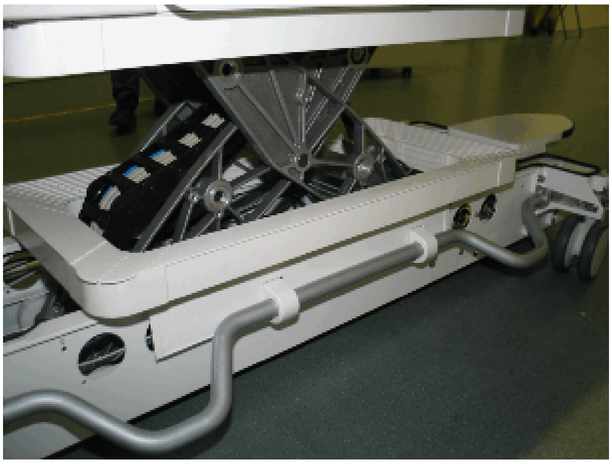


Fig. 50: Safety bolt, older scissor version

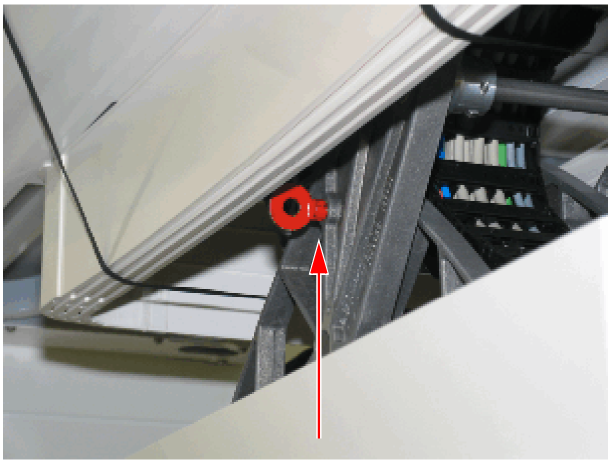
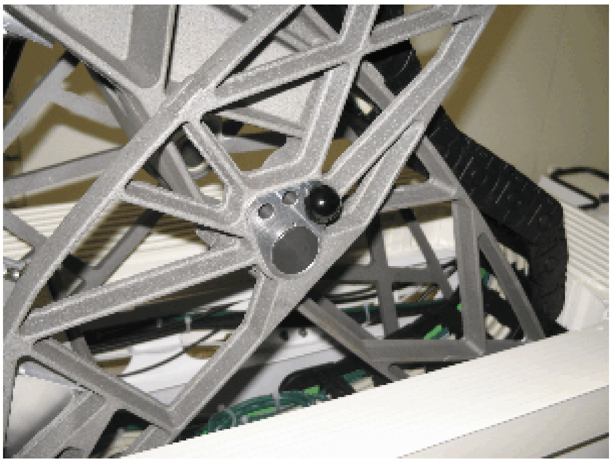
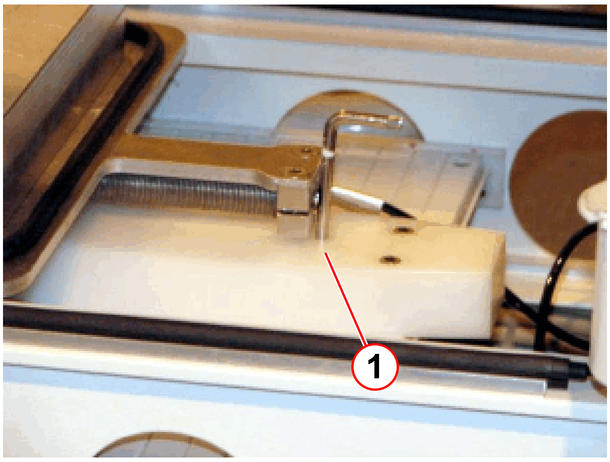


Fig. 51: Safety bolt installed, new scissor



- Remove the table column cover.
- Use the safety bolts and secure the table in the highest position.

Fig. 52: Retractable cover blocked

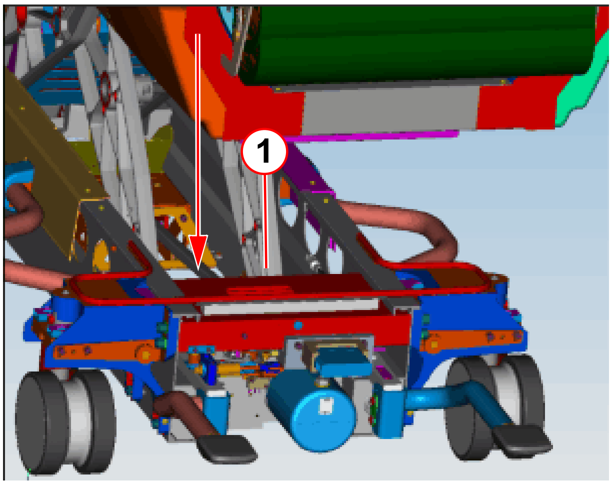


(1) 5mm allen key

- Push the retractable cover back into the table.
- Insert a **5-mm** Allen key into the hole of the plastic bracket from the retractable cover and block the cover for closing. **The retractable cover must remain in the position as shown.**
  - » There are 2 versions of plastic brackets mounted (1 hole older version,2 holes latest version)
  - » With the latest version, use the second hole from the right side of the bracket

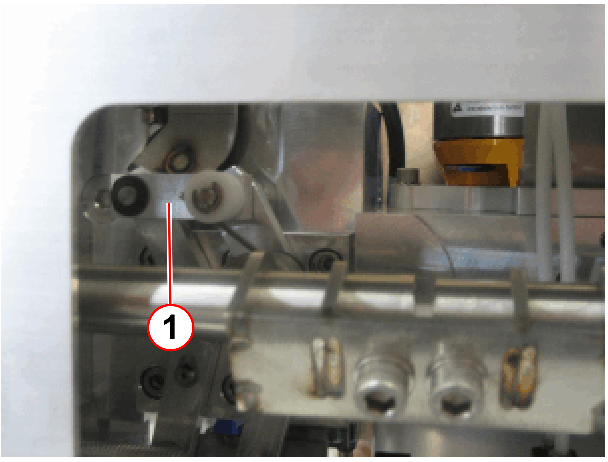
> There are 2 types of plastic brackets used for the retractable cover.
>
> Older version, 1 hole, until serial number 1028 for 3T and 1046 for 1,5T.
>
> Latest version, 2 holes, form serial number 1029 for 3T and 1047 for 1,5T.

Fig. 53: Valve position



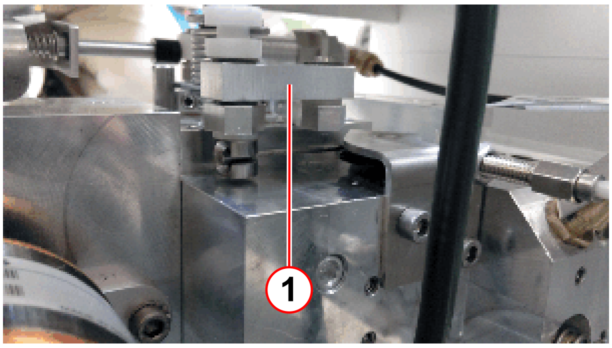
(1) View of valve position

Fig. 54: V3, V4 valve setting to the left



(1) Valve setting V3, V4 left position

Fig. 55: V3, V4 settiing to the right



(1) Valve setting V3, V4 right position (view from head end side)

- Checking actuator V3/V4, nominal position left.
  » View of valve position.

Fig. 56: Valve setting V3, V4 left setting

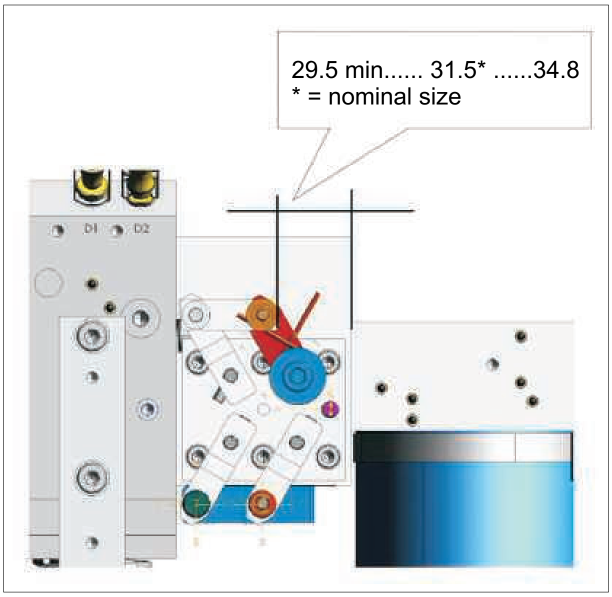
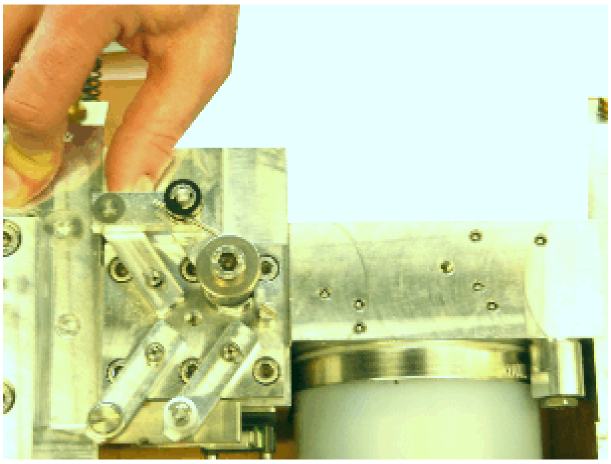
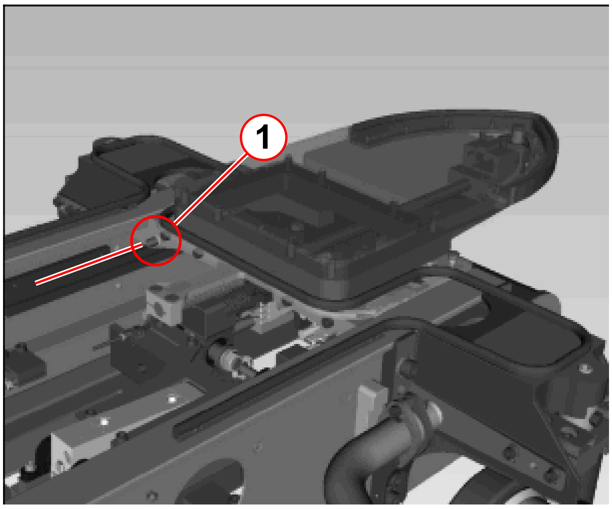


Fig. 57: Valve of lift docking switch over



- Checking actuator V3/V4, nominal position left.
  - » View of valve position.

Fig. 58: Setting for Bowden cable V3, V4



(1) Changing the depth of screw insertion

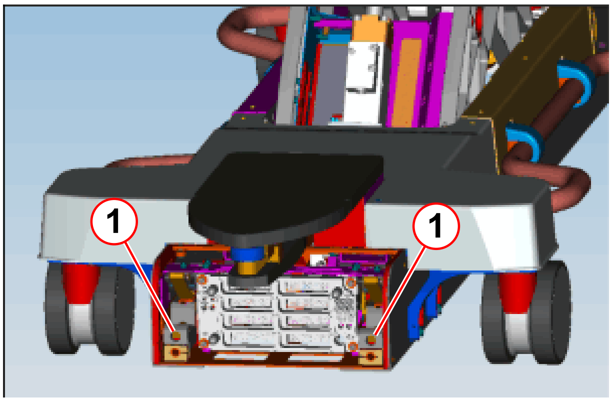
- In case of deviations, set the bowden cable length with the adjustment screw (1)

  (there is also a lock nut which need to be loosened/fastened accordingly) .

Testing

- Retractable cover is still open.

Fig. 59: Carriage of docking mechanics



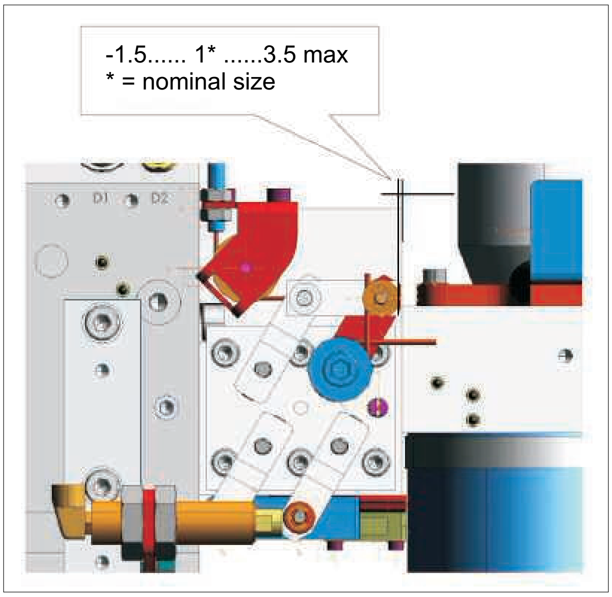
(1) Movement of carriage docking mechanics

- Activating the right hydraulic pedal (docking).
- Both carriages of the docking location travel to the rear.

- Activating the left hydraulic pedal (undocking) both carriages of the docking location travel to the front.

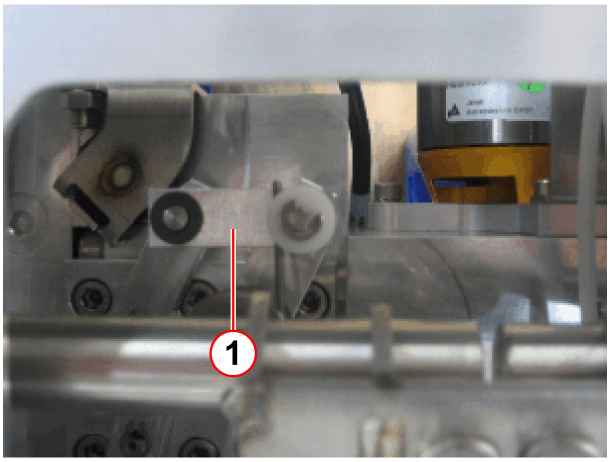
- Close retractable cover, actuators V3/V4 are located to the right as seen from the rear.

Fig. 60: Valve setting V3, V4 right setting



- Actuators V3/V4 are located to the right as seen from the rear.

Fig. 61: V3, V4 valve setting to the right



(1) Valve setting V3, V4 right position

- Remove safety bolt.
- Install covers.

5.3.1.7 Checking 5th the Wheel

Directional travel

- Switch the gear shift (lever) to the upper position - release directional travel.
- Look under the table and move it to the side -> the 5th wheel should turn to the corresponding direction and allow for lateral movement of the table.
- Set the gear shift to Center Position -> directional travel active.
- Move the table forward at least 1 m to align the 5th wheel.
- Look under the table and move it to the side -> the 5th wheel should not turn.

Releasing the 5th wheel when docking

- Shift the gear shift to the Center Position -> directional travel active.
- Move the table forward at least 1 m to align the 5th wheel.
- Bring the table toward the Docking Station at a slight angle (approx. 10 deg.) and perform the docking procedure.
- Look from the side under the table. The 5th wheel should be released and free to move away from Center Position.

Checking the pneumatic spring of the 5th wheel

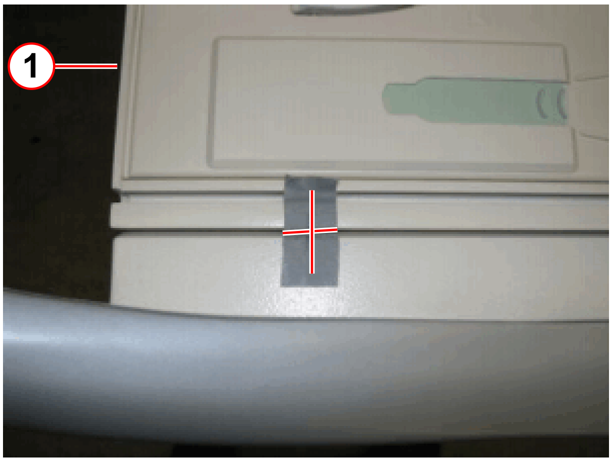
- Place both front wheels of the table on an approx. 20-mm thick base (e.g. a board).
- Grab the 5th wheel and try to push it upward.
- If you **cannot** lift the 5th wheel or it is difficult to do so, the pneumatic spring works OK.

5.3.1.8 S10 Tabletop Home Position

Table is docked

- Move table top plate electrically into the Home Position.

Fig. 62: Use adhesive tape and the mark position (food end)



(1) Foot-end side

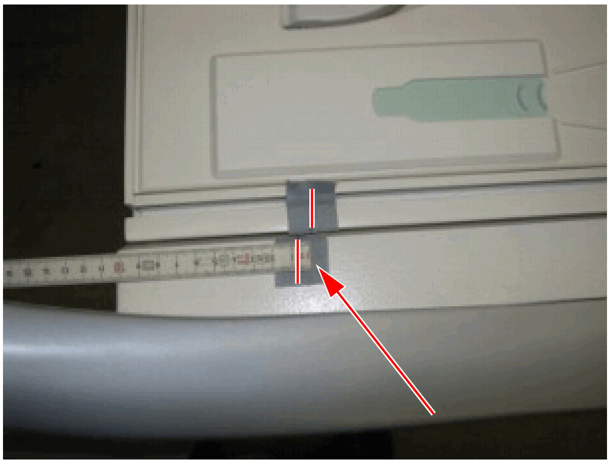
- Attach a piece of adhesive tape across table top plate and support frame.
- Draw a straight line at the tape to mark table top Home Position relative to support frame.
- Cut the adhesive tape (in the gap).

- Undock the table with the left pedal.

Fig. 63: Cut adhesive tabe and move tabletop towards

- Pull table top emergency release handle.push tabletop plate toward head-end up to end stop.
- Push table top toward head-end until the end stop (horizontal locking bolt).

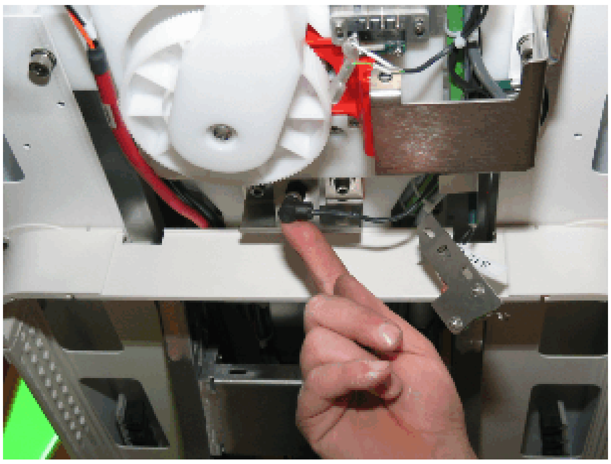
Fig. 64: Measure the difference between the marks



- Measure the difference between the marks.
  » Nominal dimensions **6 + 1 mm**.

- In case of deviations, check setting of sensor S10.

Fig. 65: Sensor S10



- Loosen the M6 screws and adjust sensor bracket with switch S10 to the correct position. Refer to:

  (→ Tune-up/Adjustments / M7-000.841.15)

5.3.1.9 Register a new DPD

> Table must be operational and docked.

A **new DPD needs to be configured** by the system Software one time:

- Open a command tool, (to read the recognized table).
- Enter the command: MrVerifyPTAB
- Enter material number and serial number with the following syntax:
  MrVerifyPTAB -m (mat) -s (ser)
  - » An **example** for a fix table with serial number **1012**.
    EXAMPLE_____MrVerifyPTAB -m 10433519 -s 1012

> Enter the material number and serial number of the complete table, not of the DPD!

Fig. 66: MrVerify-help window

```
C:\Users\meduser>mrverifyptab -help
mrverifyptab Writes the specified material and serial number to the patient table (PTAB) and MeasConfig
Usage: mrverifyptab [OPTIONS...] | [-help]
Following options are possible:
    MrVerifyPTAB                              --> Print current settings
    MrVerifyPTAB -help                        --> Show this help message
    MrVerifyPTAB -m <mat> -s <ser>            --> Write serial and material number to PTAB HW
    MrVerifyPTAB -m <mat> -s <ser> -init      --> Remove registered devices from MeasConfig,
                                                  register the new PTAB device to MeasConfig
    MrVerifyPTAB -m <mat> -s <ser> -append    --> Register the new PTAB device additional to MeasConfig
    MrVerifyPTAB -initFromPTAB                --> Remove registered PTAB devices from the system,
                                                  reads current values from the PTAB HW and
                                                  register the current PTAB HW values to MeasConfig
    MrVerifyPTAB -appendFromPTAB              --> Reads current values from the PTAB HW and
                                                  register the current PTAB HW values additional to MeasConfig
    MrVerifyPTAB -m <mat> -s <ser> -remove    --> Deregister PTAB device from MeasConfig
    MrVerifyPTAB -updateDPD                   --> Update the DPDOperatingData
                                                  (necessary after DPD-change for every PTAB HW!)
    MrVerifyPTAB -cF <factor> -correction     --> Write correction factor to PTAB HW
                                                  !!TableStop has to be pressed before enter correction factor!!
                                                  Default value for correction factor: 100150
                                                  Valid range for correction factor: 99900 - 100400

finished successfully.
```

- Reboot Meas & Recon.
- Perform TuneUp step "Table Height".

5.3.1.10 Measure and Check the Correction Factor

- Refer to REP(→ Tune-up/Adjustments / M7-000.841.15)

With the following UI`s:MR005/15/P, MR019/15/Rt, MR020/15/R, MR021/15/P, MR022/15/R, MR023/15/R the Optical length meter (OLM) and the Optical Scale (OSC) is removed and obsolete.

The correction factor from the Horizontal Gear Box must be measured, calculated and entered with the MrVerify command. The default value of the Ptab Loadware is 100150!!

5.3.1.11 Configure the Correction Factor

Table must be operational and docked. Press Table Stop before entering the correction factor.

For a **new Horizontal Drive Gear Box or a new DPD, the correction factor must be configured** by the MrVerifyPTAB tool

- Open a command tool
- Press Table Stop
- Enter the command: MrVerifyPTAB

For a new Gear Box, always check the correction factor with a folding rule. Adapt the value if necessary.

- Enter the correction factor noted on a label of the Horizontal Gear Box, after exchange of DPD:

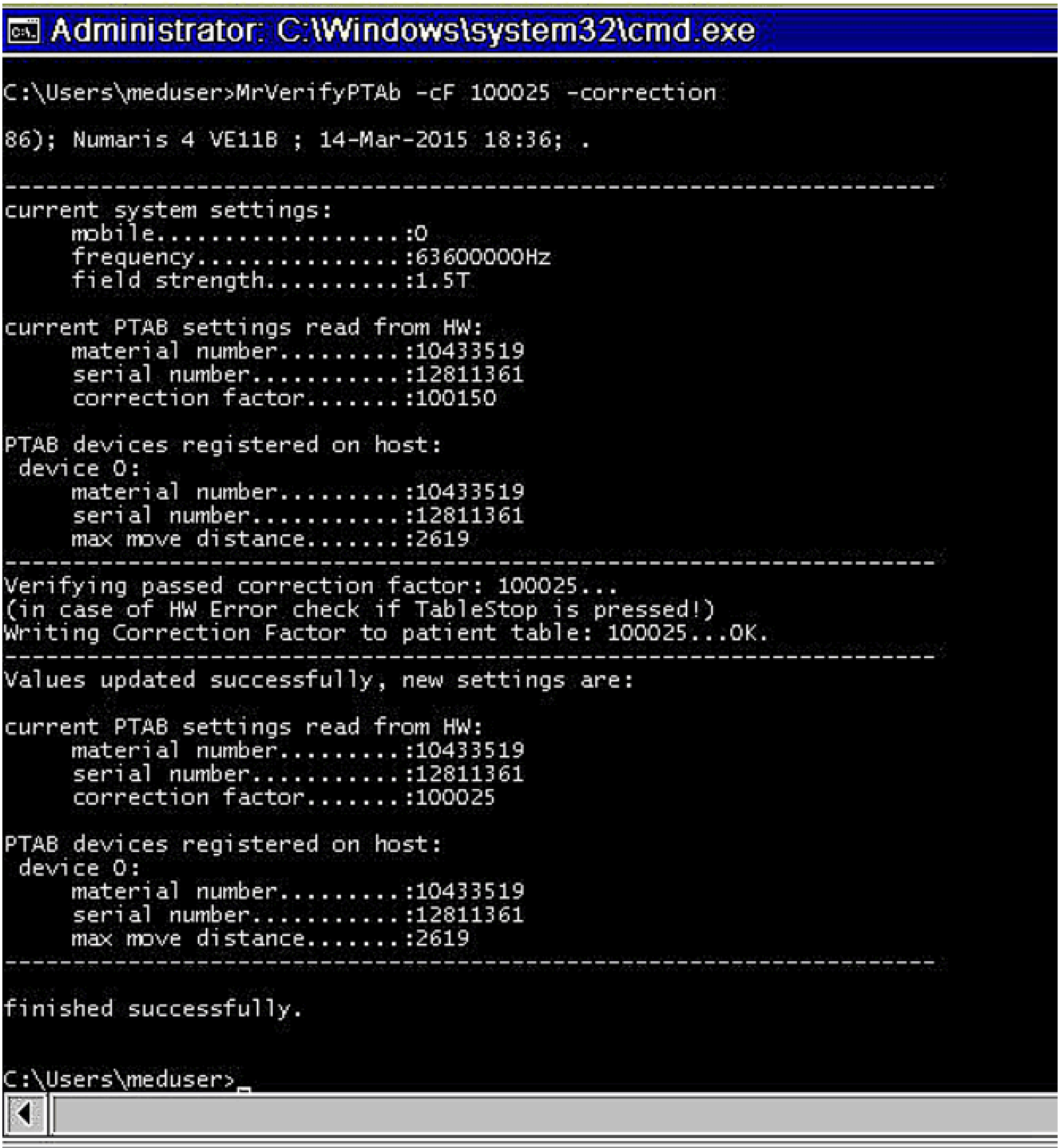
  MrVerifyPTAB -cF (factor) -correction

  » An **example** for a table with correction factor **100025**; the correction factor of the Horizontal Drive Gear Box must be entered.

  EXAMPLE_____MrVerifyPTAB -cF 100025 -correction

Enter the correction factor of the label from the Horizontal Gear Box, the default value of correction factor in the Ptab Loadware is 100150!!

Fig. 67: Correction factor

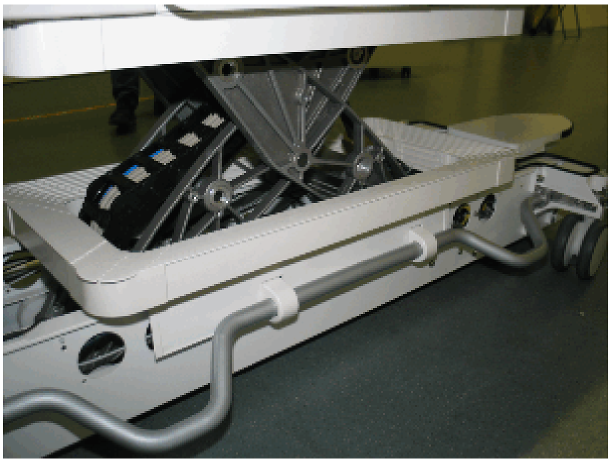


```
Administrator: C:\Windows\system32\cmd.exe

C:\Users\meduser>MrVerifyPTAb -cF 100025 -correction

86); Numaris 4 VE11B ; 14-Mar-2015 18:36; .

----------------------------------------------------------------------
current system settings:
     mobile..................:0
     frequency...............:63600000Hz
     field strength..........:1.5T

current PTAB settings read from HW:
     material number.........:10433519
     serial number...........:12811361
     correction factor.......:100150

PTAB devices registered on host:
 device 0:
     material number.........:10433519
     serial number...........:12811361
     max move distance.......:2619
----------------------------------------------------------------------
Verifying passed correction factor: 100025...
(in case of HW Error check if TableStop is pressed!)
Writing Correction Factor to patient table: 100025...OK.
----------------------------------------------------------------------
Values updated successfully, new settings are:

current PTAB settings read from HW:
     material number.........:10433519
     serial number...........:12811361
     correction factor.......:100025

PTAB devices registered on host:
 device 0:
     material number.........:10433519
     serial number...........:12811361
     max move distance.......:2619
----------------------------------------------------------------------

finished successfully.

C:\Users\meduser>_
```

- Install covers.
- Release Table Stop.
- Reboot Meas & Recon.

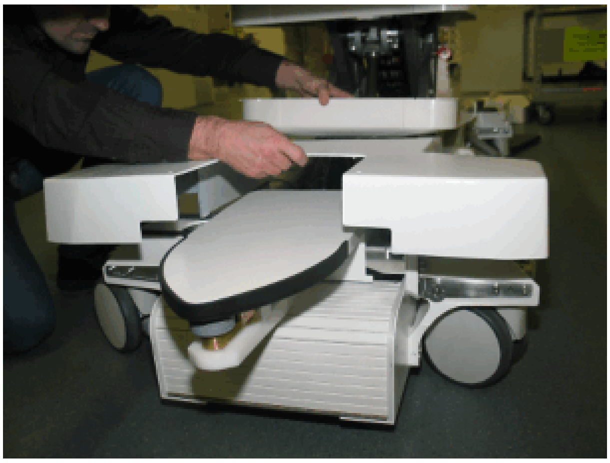
5.3.1.12 Checking for Damage of Bowden Cables

Fig. 68: Table column cover



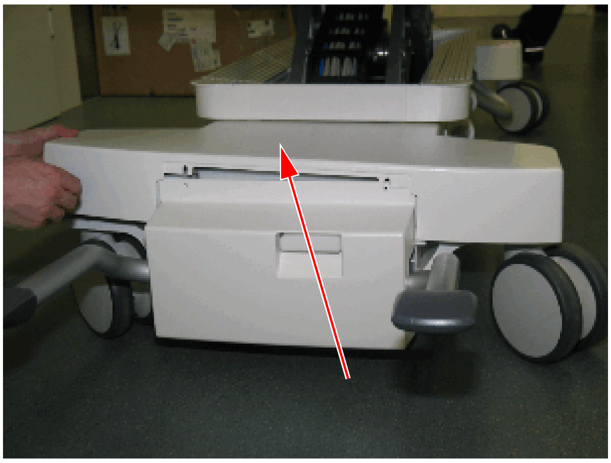
- Remove the table column cover.

Fig. 69: Chassis cover head end



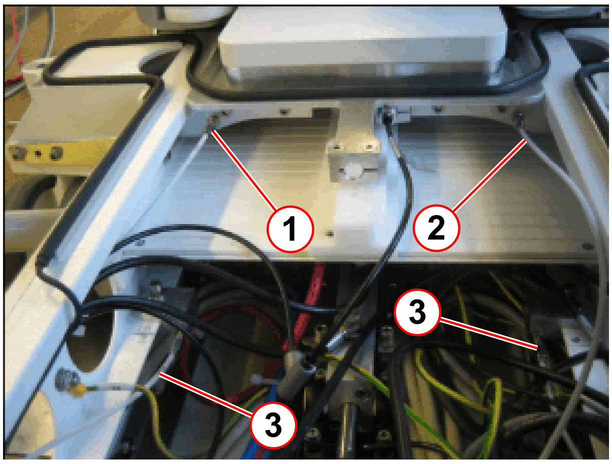
- Remove the screws of the cover chassis head-end.

Fig. 70: Cover chassis - foot end top



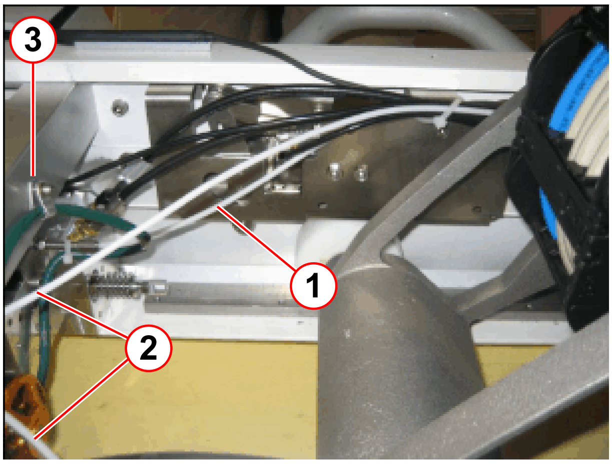
- Remove the cover chassis of the foot end top.

Fig. 71: Overview hydraulic hoses and bowden cables



(1) Bowden cable switch over lift/dock
(2) Bowden cable switch over 5th wheel
(3) Bowden cables undock emergency release

Fig. 72: Overview bowden cables switch over lift/dock and emergency release



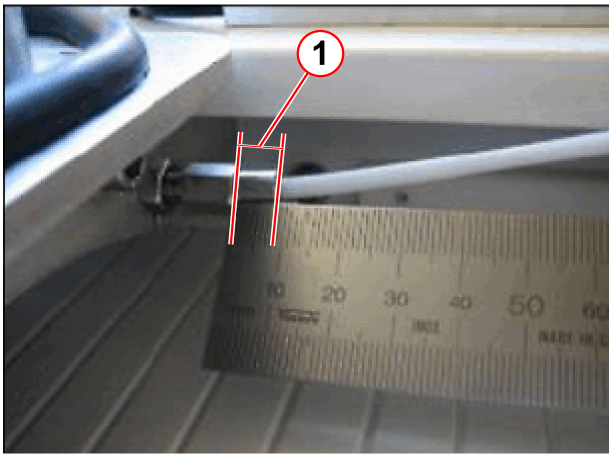
(1) Bowden cable switch over lift/dock
(2) Bowden cables undock emergency release
(3) Head end side

- Check the hoses of the bowden cable for damage.
- Pull on the hoses of the bowden cable and check if the bowden cable is in good order.

5.3.1.13 Checking the (Gear Shift) Bowden Cable Friction of the 5th Wheel

- Roller brake is in Center Position.

Fig. 73: Bowden cable clearance



(1) Max. clearance of bowden cable cover 6-8 mm

- Turn the adjustment screw of the bowden cable fully in.
- Slightly pull the hose of the bowden cable, and measure
  - » the distance from the end fitting to the adjustment screw must be adjusted to approx. 6-8 mm.

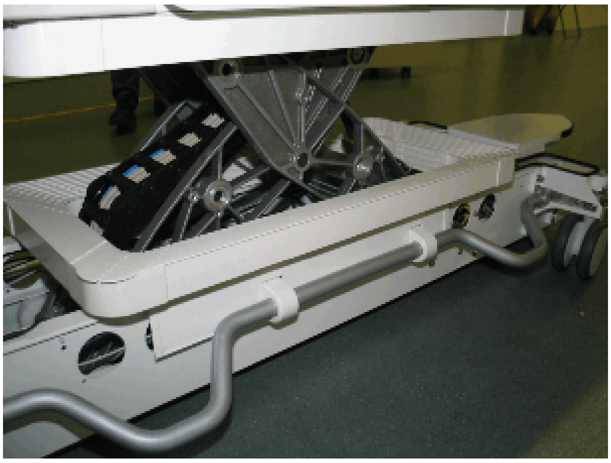
- Install covers.

5.3.1.14 Checking the Push Crank Mechanism

If docking is difficult

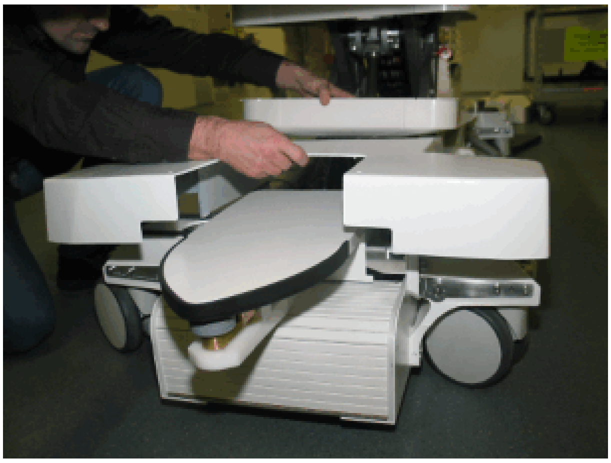
- Remove the cover from the Docking Station.

Fig. 74: Table column cover



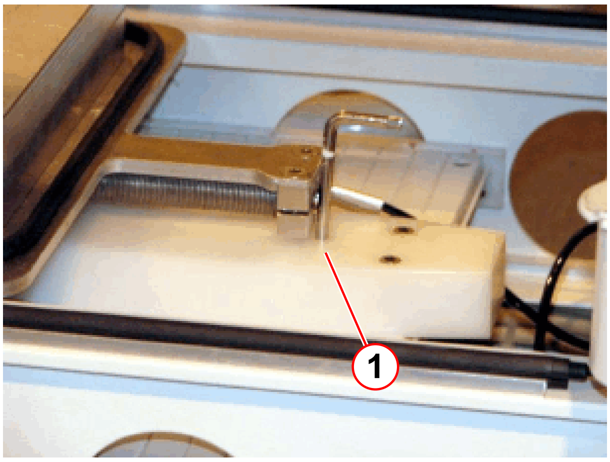
- Remove the table column cover.

Fig. 75: Chassis cover head end



- Remove the screws of the cover chassis head-end.

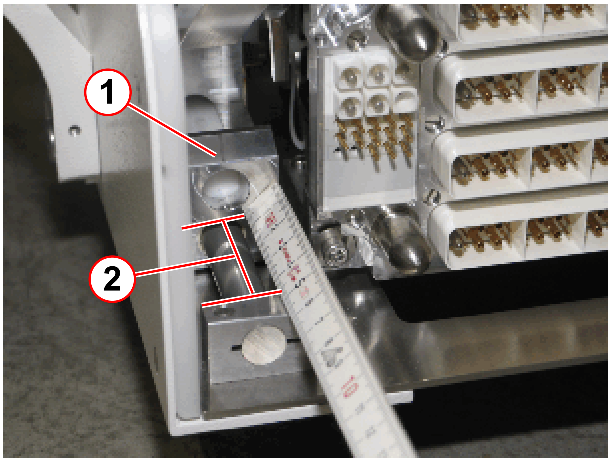
Fig. 76: Retractable cover blocked



(1) 5mm allen key

- Push the retractable cover back into the table.
- Insert a 5-mm Allen key into the hole of the plastic bracket from the retractable cover and block the cover for closing.

Fig. 77: Push crank mechanics



(1) Hydraulic pusher
(2) Distance between carriage and inner side of mounting bracket

- A second person should push the right docking pedal.
- The hydraulic pusher (1) slides to the back.
- Push the left undock pedal. The hydraulic pusher slides back to front.
- Distance (2) from the mounting bracket to the lower hydraulic pusher must be a minimum of **37 mm**.

If the hydraulic pusher does not move:

- Emergency release is activated. Check bowden cables of emergency undock mechanics.
    - » Reset emergency release
- S50 not sufficiently activated.
    - » Check adjustment S50
- Check adjustment of Lift-Docking Switch-Over.
- Install covers.

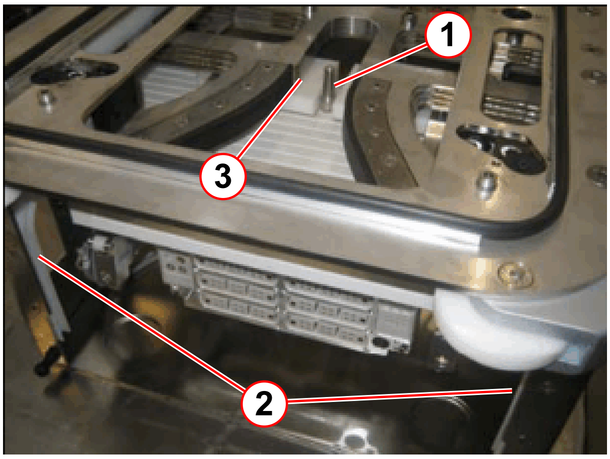
5.3.1.15 Checking the Docking Station Retractable Cover Mechanism

If movement is difficult:

Table is undocked

- Remove the cover from the Docking Station.

Fig. 78: Retractable cover guide



(1) Positioning an item
(2) Retractable cover guides
(3) Plastic actuator

- Manually open the retractable cover and keep it locked in open position by positioning an item (1), for example a piece of wood, between the wedges. Then lubricate the retractable cover guides (2) and the plastic actuators (3) with silicone spray.
- Install covers.

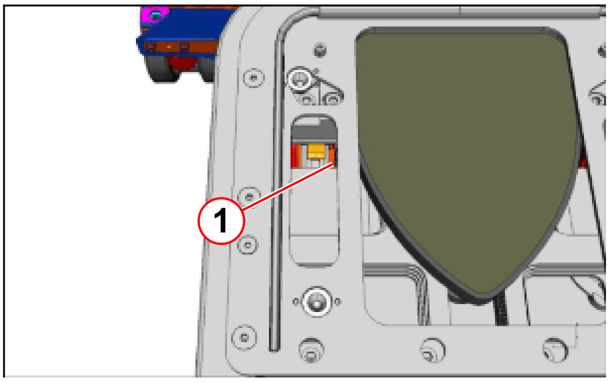
5.3.1.16 Docking Mechanics Latches

Table is docked

- The table cannot be undocked because the white plastic latches stick.
- Remove the Docking Station cover.
- Manually lift the white latches (1) upward using a long screwdriver.

- The table now can be pushed away from the Docking Station.

Fig. 79: Docking mechanics latches



(1) Latches

- The white plastic latches should be replaced.

- Install covers.

5.4 Configuration

- A **new DPD needs to be configured** by the system Software one time, see (→ Register a new DPD / Page 97)

6.1 Light Localizer - Quick Alignment Check

6.1.1 Requirements

Prerequisite: The patient table is correctly aligned horizontally.

6.1.1.1 Estimated Completion Time and Number of Persons

Tab. 14 PDD

Work Steps	Time	Persons
Check Light Localizer	5 min	1

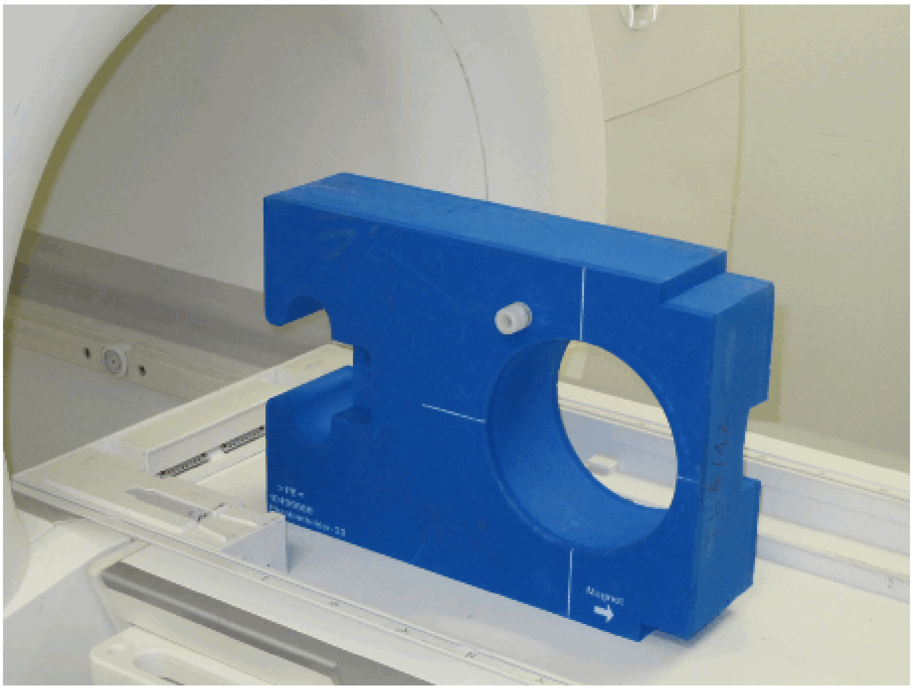
6.1.1.2 Aids or Tools

- Phantom holder 23 (material number 104 96 688 - blue phantom holder for 240 mm spherical phantom)

6.1.2 Procedure

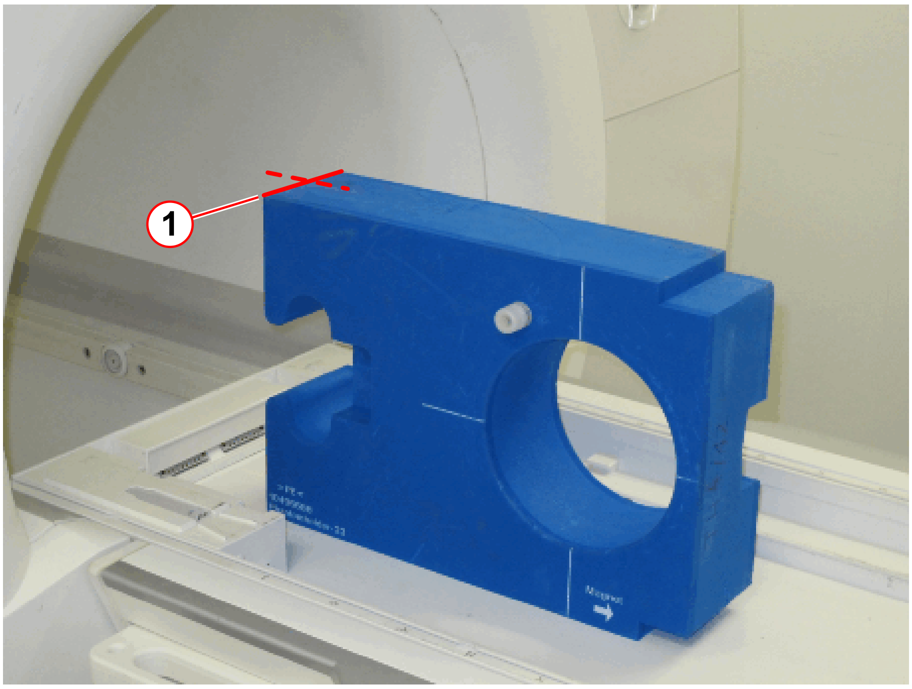
1. Place phantom holder 23 upright.

Fig. 80: Phantom Holder - upright position



2. **Align the laser to the upper front edge** of the phantom holder.

Fig. 81: Phantom Holder - light localizer

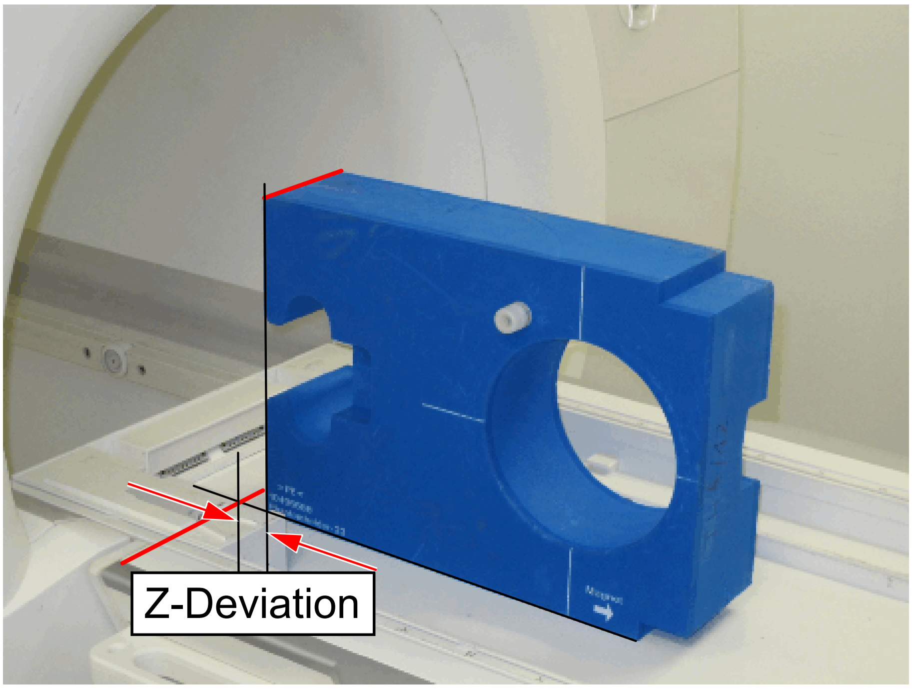


(1) Align Laser

3. At the **lower phantom edge**, measure the laser Z-deviation.

   » Note down the value (e.g. +2 mm).

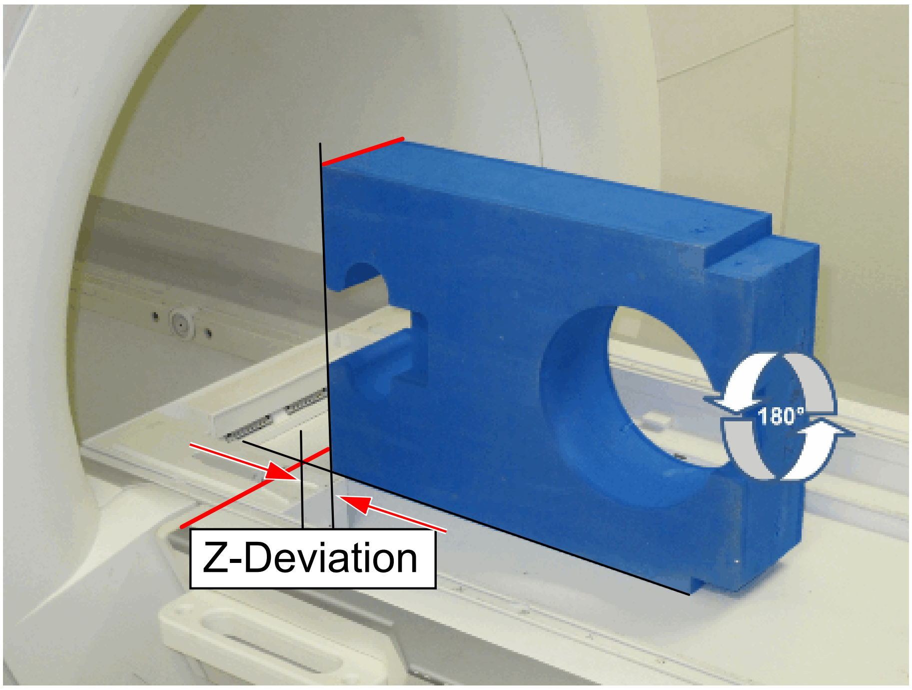
Fig. 82: Z-Deviation of laser



4. **Ensure the phantom holder is orthogonal**. Rotate the holder by 180 degrees. Measure Z-deviation again.

   » The **Z-deviation must be identical as before** (e.g. +2 mm, see step 3. before).

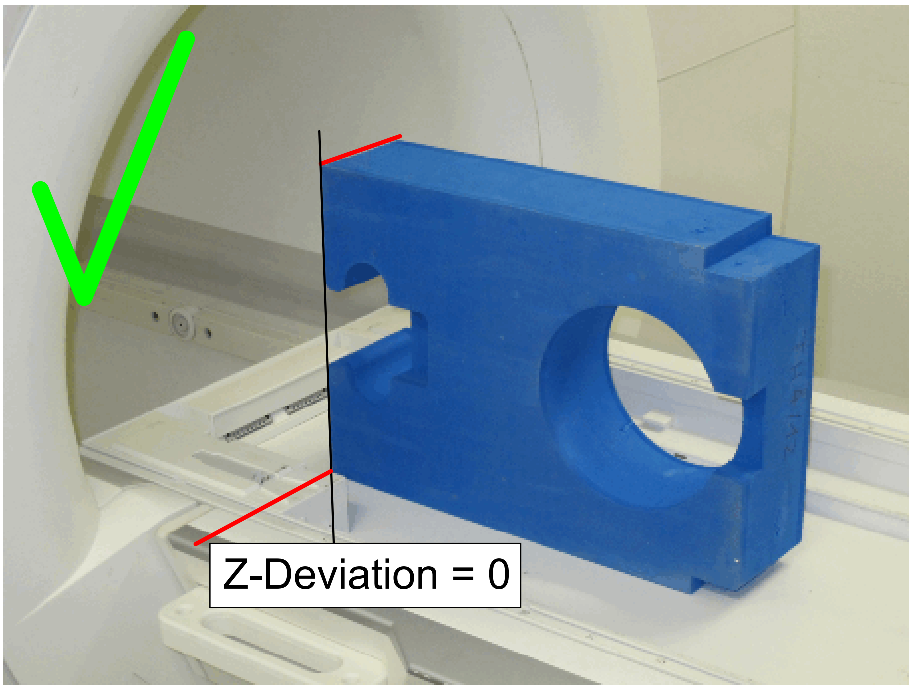
Fig. 83: Z-Deviation with Phantom Holder rotated 180 degrees



5. In case the measured **Z-Deviation was more than 1 mm, adjust the Light Localizer**, see Replacement of Parts / Patient Handling (→ Setting the laser light localizer / M7-000.841.15).
6. After adjustment, verify correct light localizer adjustment.

   » **The Z-Deviation must be 0 mm**.

Fig. 84: Z-Deviation correctly adjusted (0 mm)



7. Perform TuneUp / QA:
   - TU / Light Marker
   - all other dependant TU steps which are toggled to “To Do” status.
   - all dependant QA steps which are toggled to “To Do” status.

6.2 Light Localizer - Adjustment Procedure

see Replacement of Parts / Patient Handling (→ Setting the laser light localizer / M7-000.841.15).

Version 06:

- Adjustment procedures S12/S50 updated.
- Pictures has been updated.
- Editorial changes.

Version 07:

- **The following systems were added:**
  - Avanto^{fi}t
  - Skyra^{fit}
  - Prisma
  - Prisma^{fit}
- Editorial changes.
- (→ Hints for Troubleshooting / Page 62) updated.

Version 08:

- (→ Hints for Troubleshooting / Page 62) updated, troubleshooting procedure for Table Stop added.
- Chapter “Hints for Troubleshooting” “Configure the correction factor” added. (→ Configure the Correction Factor / Page 98)
- Chapter “Hints for Troubleshooting” “Register a new DPD” revised.
- Editorial changes.

Version 09:

- Editorial changes.

Version 10:

- New Chapter (Cabin User Interface) - quick check for correct light localizer alignment (→ Light Localizer - Quick Alignment Check / Page 107).
- Chapter 5.3 Hints for Troubleshooting revised, “Table undocked”.
- All chapters - minor editorial changes.
- Chapter patient table - troubleshooting strategy referenz to error message changed from separat document to SKB.
- New pictures added..

There are no Hazard IDs in this document.

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